



**CLEAN DEVELOPMENT MECHANISM
PROJECT DESIGN DOCUMENT FORM (CDM-PDD)
Version 03 - in effect as of: 28 July 2006**

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**SECTION A. General description of project activity****A.1. Title of the project activity:**

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Qingdao Longxin Wind Power Project Phase I

Version number of the document: 03

Date: 18/04/2012

A.2. Description of the project activity:

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Qingdao Longxin Wind Power Project Phase I (hereinafter referred to as the proposed project) is to utilize wind resources for electricity generation through the construction of a wind farm with a total capacity of 49.8MW in Liyuan Town and Dianzi town, Pingdu City, Shandong Province, P. R. China. The electricity generated from the project will be sold to North China Power Grid (NCPG). The proposed project will achieve obvious greenhouse gas (GHG) emission reductions through the displacement of electricity delivered by North China Power Grid which is a fossil-fuel dominated grid.

The purpose of the proposed project is to generate zero-emission wind power and deliver it to North China Power Grid. For the proposed project,

- (a) Prior to the start of implementation of the project activity, there is no power generation unit at the site of the proposed project, and the electricity was supplied by the North China Power Grid which was dominated by fossil fuel-fired power plants.
- (b) The project scenario is the implementation of the proposed project, the installation of 24 wind turbines with capacity of 2000 kW each and 1 wind turbine with capacity of 1800 kW with a total capacity of 49.8MW will supply an average annual generation of 99,026 MWh to North China Power Grid and replace the same amount of electricity generated by fossil fuel-fired power plants connected to North China Power Grid. According to ACM0002 applied, the proposed project is a zero-emission electricity generating activity.
- (c) The baseline scenario of the proposed project is the electricity supply of equal amount as the proposed project from the North China Power Grid. The baseline scenario of the proposed project is the same as the scenario prior to the start of the implementation of the project activity.

The proposed project involves the installation of 24 wind turbines with capacity of 2000 kW each and 1 wind turbines with capacity of 1800 kW which amount to a total installed capacity of 49.8MW. The proposed project is constructed and operated by China Resources Wind Power (Qingdao) Co., Ltd. The estimated annual net electricity generation and average annual emission reductions of the proposed project are 99,026MWh and 92,183tCO₂e, respectively.

The proposed project makes contribution to the local sustainable development as follows:

1. GHG emission reduction

The proposed project activity will achieve obvious greenhouse gas (GHG) emission reductions by avoiding CO₂ emissions, as grid-connected fossil fuel-fired power dominates in the North China Power Grid.

**2. Pollutants emission reduction through replacing fossil fuel combustion**

The proposed project is to replace grid-connected fossil fuel-fired power plants in the North China Power Grid, and thus reduce fossil fuel consumption and avoid pollutants emission, such as sulfur dioxide and dust, brought by fossil fuel combustion. Therefore, the proposed project has obvious environmental benefit.

3. Employment opportunities

The conducting of the proposed project will offer 20 job opportunities for local people during both the construction and operational period.

4. Economy development

The region can achieve economic growth through the construction and operation of the project. Furthermore, the project will contribute to local government with more tax revenues and poverty eradication.

A.3. Project participants:

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Please list project participants and Party(ies) involved and provide contact information in Annex 1. Information shall be indicated using the following tabular format.

Name of Party involved (*) (host) indicates a host Party)	Private and/or public entity(ies) project participants (as applicable)	Kindly indicate if the Party involved wishes to be considered as project participant (Yes/No)
People's Republic of China (host)	China Resources Wind Power (Qingdao) Co., Ltd.	Yes

(*) In accordance with the CDM modalities and procedures, at the time of making the CDM-PDD public at the stage of validation, a Party involved may or may not have provided its approval. At the time of requesting registration, the approval by the Party (ies) involved is required.

Further contact information of project participants is provided in Annex 1.

A.4. Technical description of the project activity:**A.4.1. Location of the project activity:****A.4.1.1. Host Party(ies):**

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People's Republic of China

A.4.1.2. Region/State/Province etc.:

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Shandong Province

A.4.1.3. City/Town/Community etc:

Liyuan Town and Dianzi town, Pingdu City

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The proposed project is located at Liyuan Town and Dianzi Town, Pingdu City, Shandong Province, P. R. China. The nearest Liyuan Town is about 10 kilometers from the project site. The geographical coordinates of the proposed project is from 119.8827°E to 119.9503°E, and from 36.8157°N to 37.8915°N (The central GPS coordinate is 119.91125°E, 36.8568°N). Figure A1 and A2 show the geographical location of the proposed project.

Figure A1. The proposed project on the map of P. R. China



Figure A2. The proposed project on the map of Pingdu City, Shandong Province, P. R. China



A.4.2. Category(ies) of project activity:

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Category: Renewable electricity in grid-connected applications

Sectoral Scope: 1 Energy industries

A.4.3. Technology to be employed by the project activity:

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The proposed project is to utilize wind resources for electricity generation in Pingdu City, Shandong Province, P. R. China. The proposed project is a grid-connected renewable energy project.

Prior to the start of implementation of the project activity, there is no power generation unit at the site of the proposed project, and the electricity was supplied by the North China Power Grid. The baseline scenario of the proposed project is the electricity supply of equal amount as the proposed project from the North China Power Grid. In the North China Power Grid, installed thermal power plants account for 92.84% of the total installed capacity in 2009; hydropower plants account for 3.00%; wind power plants and other kinds of power plants account for 4.16%¹.

The project scenario is the implementation of the proposed project, the installation and operation of 24 wind turbines with capacity of 2000 kW each and 1 wind turbines with capacity of 1800 kW with a total capacity of 49.8MW which will supply an average annual generation of 99,026 MWh to North China Power Grid and replace the same amount of electricity generated by fossil fuel-fired power plants connected to North China Power Grid. According to ACM0002 (version 12.2.0) applied, the proposed project is a zero-emission electricity generating activity.

¹ China Electric Power Yearbook 2010



The baseline scenario of the proposed project is the same as the scenario prior to the start of the implementation of the project activity.

By replacing the electricity generated from fossil fuel-fired power plants dominated North China Power Grid; the proposed project activity will achieve obvious greenhouse gas (GHG) emission reductions by avoiding CO₂ emissions. According to ACM0002 (version 12.2.0) applied, the proposed project is a zero-emission electricity generating activity.

The proposed project involves the installation of 24 wind turbines with capacity of 2000 kW each and 1 wind turbine with capacity of 1800 kW, which amount to a total installed capacity of 49.8 MW. The estimated power output is 99,026MWh and the load factor is 22.70%. The main technical specifications of the wind turbine are provided in the following table.

Item	Unit	Index	Index
Model		V90-2.0MW	V90-1.8MW
Rated capacity	kW	2000	1800
Number of blades		3	3
Rotor diameter	m	90	90
Rated wind speed	m/s	13	12
Shutoff speed	m/s	2.5m/s	3.5m/s
Height of hub	m	80	80
Rated voltage of generator	V	690	690

An 110kV substation will be built at the proposed project site, which connects the proposed project to the Tangtian 220kV Substation which is about 6.7 km away from the proposed project and belongs to the North China Power Grid. Then the power produced in the wind farm can be transmitted to the North China Power Grid. Grid-connected electricity generated by the proposed project will be monitored through metering equipment (meter monitoring both electricity input and electricity output) at the Tangtian 220kV substation. The data can also be monitored and recorded at the on-site control centre using a computer system.

The proposed project will use wind turbines manufactures by foreign supplier, therefore there is international technology transfer involved

A.4.4. Estimated amount of emission reductions over the chosen crediting period:

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A crediting period of 7 years (20/07/2012-19/07/2019, renewable twice) is selected for the project activity. An estimation of emissions reductions expected over the crediting period is provided in the table below.

Years	Annual estimation of emission reductions in tonnes of CO ₂ e
20/07/2012-19/07/2013	92,183
20/07/2013-19/07/2014	92,183
20/07/2014-19/07/2015	92,183
20/07/2015-19/07/2016	92,183
20/07/2016-19/07/2017	92,183



20/07/2017-19/07/2018	92,183
20/07/2018-19/07/2019	92,183
Total estimated reductions (tonnes of CO ₂ e)	645,281
Total number of the first crediting years	7
Annual average over the crediting period of estimated reductions (tonnes of CO ₂ e)	92,183

A.4.5. Public funding of the project activity:

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There is no public funding for this project.

**SECTION B. Application of a baseline and monitoring methodology****B.1. Title and reference of the approved baseline and monitoring methodology applied to the project activity:**

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The approved methodology applied in the proposed project activity is ACM0002 (version 12.2.0) – “Consolidated methodology for grid-connected electricity generation from renewable sources”.

Reference:

<http://cdm.unfccc.int/methodologies/DB/C505BVV9P8VSNNV3LTK1BP3OR24Y5L>

The following tools are also applied in the proposed project:

“Tool for the Demonstration and Assessment of Additionality (version 05.2.1)”

Reference:

<http://cdm.unfccc.int/methodologies/PAmethodologies/tools/am-tool-01-v5.2.1.pdf>

“Tool to calculate the emission factor for an electricity system (version 02.2.0)”

Reference:

<http://cdm.unfccc.int/methodologies/PAmethodologies/tools/am-tool-07-v2.2.0.pdf>

For more information regarding the methodologies please refer to

<http://cdm.unfccc.int/methodologies/PAmethodologies/approved.html>

B.2 Justification of the choice of the methodology and why it is applicable to the project activity:

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The proposed project installs a new power plant at a site where no renewable power was operated prior to the implementation of the project activity. It meets all applicability conditions of methodology ACM0002 (version 12.2.0) which is listed as follows:

- 1) The proposed project is a grid-connected renewable new wind power project; and it does not involve capacity addition, retrofit or replacement;
- 2) The proposed project does not involve switching from fossil fuels to renewable energy at the site.
- 3) The geographic and system boundaries of North China Power Grid to which the proposed project will be connected can be clearly identified and information on the characteristics of the grid is available.

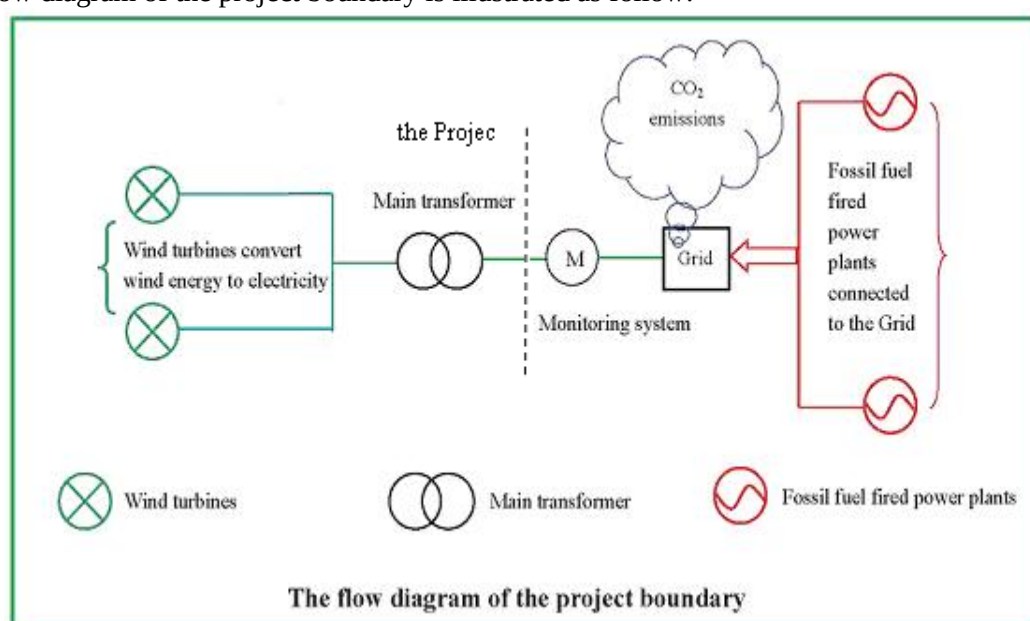
B.3. Description of the sources and gases included in the project boundary

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	Source	Gas	Included	Justification / Explanation
Baseline	Fossil fuel-fired power plants in the North China Power Grid	CO ₂	Yes	Major emission sources.
		CH ₄	No	Minor emission source.
		N ₂ O	No	Minor emission source.

Project Activity	The proposed wind power project	CO ₂	No	According to methodology, the proposed project is a wind power project, so it does not involve project emission.
		CH ₄	No	According to methodology, the proposed project is a wind power project, so it does not involve project emission.
		N ₂ O	No	According to methodology, the proposed project is a wind power project, so it does not involve project emission.

The flow diagram of the project boundary is illustrated as follow:



The spatial extent of the proposed project boundary includes the proposed project site and all power plants connected physically to the North China Power Grid (NCPG).

According to the “Tool to calculate the emission factor for an electricity system” (version 02.2.0), the delineation of grid boundaries as provided by the DNA of China is used². North China Power Grid composed of Beijing, Tianjin, Hebei, Shanxi, Shandong and Inner Mongolia Autonomous Region Power Grids is the project electricity system.

B.4. Description of how the baseline scenario is identified and description of the identified baseline scenario:

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According to ACM0002 (version 12.2.0), if the project activity is the installation of a new grid-connected renewable power plant/unit, the baseline is the following:

Electricity delivered to the grid by the project activity would have otherwise been generated by the

² Notification on the publication of Emission Factor of Regional Power Grid in China of Year 2010
<http://cdm.ccchina.gov.cn/WebSite/CDM/UpFile/File2552.pdf>



operation of grid-connected power plants and by the addition of new generation sources, as reflected in the combined margin (CM) calculations described in the “Tool to calculate the emission factor for an electricity system”.

Parameters used to determine the baseline emission are listed in the following table:

Parameter	Data	Source
$EF_{grid,OM,y}$ (tCO ₂ /MWh)	0.9914	Calculated in Annex 3
$EF_{grid,BM,y}$ (tCO ₂ /MWh)	0.7495	Calculated in Annex 3
$EF_{grid,CM,y}$ (tCO ₂ /MWh)	0.9309	Calculated in Annex 3

Therefore, supply of equivalent annual power output by the North China Power Grid is the baseline scenario for the proposed project activity.

B.5. Description of how the anthropogenic emissions of GHG by sources are reduced below those that would have occurred in the absence of the registered CDM project activity (assessment and demonstration of additionality):

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The proposed project has taken CDM support into account to overcome potential financial barrier since feasibility study. The Feasibility Study Report gave detailed discussions about improving the risk-resistance capability of the proposed project by applying for CDM.

Prior consideration of the CDM

Time	Project action
09/2010	Environmental Impact Assessment (EIA) developed
12/10/2010	EIA was approved by Qingdao Environmental Protection Bureau
28/12/2010	Feasibility Study Report (FSR) developed by Shandong Industrial Design Institute Co., Ltd.
02/03/2011	Feasibility Study Report (FSR) approved, in which applying for CDM support to enhance the financial attractiveness is introduced to the project developer.
15/03/2011	Board decision on the CDM development was made
20/03/2011	Stakeholders' meeting held
30/03/2011	Turbines Purchasing Agreement signed
19/05/2011	The construction contract signed
26/05/2011	Construction started
25/04/2011	CDM Consultancy Agreement signed
19/09/2011	Prior consideration of CDM was confirmed by Chinese DNA
17/10/2011	Prior consideration of CDM form was confirmed to be received by UNFCCC secretariat

Based on the FSR of the proposed project, the project developer found the Project IRR is much lower than the benchmark of 8% without revenue from CERs. The proposed project is thus financially unattractive to investors. Then the project owner, China Resources Wind Power (Qingdao) Co., Ltd.,



decided to seek additional financial support from CDM to make the proposed project feasible on the Board meeting and made investment decision on 15th Mar 2011. The proposed project started on 30th March 2011 which the date of turbines purchasing agreement signed.

The Notification of Prior Consideration of the CDM Form was approved by UNFCCC secretariat on 17/10/2011 and approved by NDRC on 19/09/2011, due to the starting date of the proposed project is 30/03/2011, therefore, the notifications of Prior Consideration of CDM were made within six months in line with “Guideline on the Demonstration and Assessment of Prior CDM Consideration (version 04)”. In the Prior Consideration of the CDM Form, precise geographical location and a brief description of the proposed project was included.

Therefore, the development procedure shows that CDM was seriously considered before the proposed project started.

The additionality of the proposed project is demonstrated and assessed by the approved “Tool for the Demonstration and Assessment of Additionality” (Version 05.2.1). Following steps include:

Step 1. Identification of alternatives to the project activity consistent with current laws and regulations

Due to the proposed project is an installation of a new grid-connected renewable power plant, and is not a capacity addition, retrofit or replacement of existing grid-connected renewable power plant/unit, the baseline scenario, according to the methodology of ACM0002 (version 12.2.0), is the following:

Electricity delivered to the grid by the project activity would have otherwise been generated by the operation of grid-connected power plants and by the addition of new generation sources, as reflected in the combined margin (CM) calculations described in the Tool to calculate the emission factor for an electricity system.

According to the ACM0002 (Version 12.2.0) and relevant requirements of VVM, alternatives to the proposed project is not needed to be identified as the baseline scenario has been prescribed in the ACM0002 (Version 12.2.0) applied.

Step 2. Investment analysis

The purpose of this step is to determine whether the proposed project activity is economically or financially less attractive than other alternatives without an additional funding that may be derived from the CDM project activities. The investment analysis was conducted in the following steps:

Sub-step 2a. Determine appropriate analysis method

The three analysis methods suggested by *Tools for the demonstration and assessment of additionality* (version 05.2.1) are simple cost analysis (Option I), investment comparison analysis (Option II) and benchmark analysis (Option III). Since the proposed project will earn revenues from not only the CDM but also the electricity output, the simple cost analysis method is not appropriate. Investment comparative analysis method is only applicable to the case that alternative baseline scenario is similar to the proposed projects, so that comparative analysis can be conducted. The baseline scenario of the proposed project is to supply equivalent annual power output from the North China Power Grid rather than a new investment project. Therefore, Option II is not an appropriate method either. The proposed project will use benchmark analysis method based on total investment IRR.

Sub-step 2b. Apply benchmark analysis (Option III)



With reference to the *Interim Rules on Economic Assessment of Electric Engineering Retrofit Projects*, issued by the State Power Corporation in 2002, the financial benchmark return adopted as a guideline for investments in the electric power industry in China is 8% of total investment for the Project IRR and only if the Project IRR of the project is higher than or equivalent to this benchmark, the proposed project is financially feasible to investors. Therefore, 8% is adopted as the relevant benchmark. The authoritative reference is deemed an appropriate benchmark reference for the retrofit power projects and new power projects investment in China because of the degree of relevance to the power industry. In section 1.11 of the reference document, it indicates that the economic benchmark is regulated for the entire power industry. The proposed project's FSR also adopts the 8% Project IRR for financial analysis. Developed by an accredited entity in China, the FSR was also approved by the local authority. As the wind farm technology applied is a mature new technology available in the wind turbine market, and thus it is applicable to use this benchmark of 8%. In addition, this benchmark is also commonly applied in China's CDM projects for the benchmark of power industry.

Sub-step 2c. Calculation and comparison of financial indicators

Based on the above-mentioned benchmark, the calculation and comparative analysis of financial indicators for the proposed project are carried out in sub-step 2c.

(1) Basic parameters for calculation of financial indicators

Based on the Feasibility Study Report of the proposed project, basic parameters for calculation of financial indicators are as follows:

<i>Parameter</i>	<i>Data</i>	<i>Data Source</i>
Installed capacity	49.8 MW	FSR,Chapter 1-Page5
Net power supply	99,026 MWh	FSR,Chapter 1-Page5
Project lifetime	21 years (one year for construction and twenty years of operation)	FSR,Chapter 1-Page10
Total static investment	528.86 million RMB	FSR,Chapter 17-Page199
Annual O&M cost	11.84 million RMB (average value)	FSR,Chapter 17-Page200
Depreciation period	15 years	FSR,Chapter 17-Page193
Current asset	1.49 million RMB	FSR,Chapter 17-Page193
Tariff	0.61 RMB/kWh (including VAT)	Fagaijiage [2009]1906
Residual value	5%	FSR,Chapter 17-Page193
Tax rate	17% ³ (Value added tax)	FSR,Chapter 17-Page194
	25% ⁴ (Income tax)	

³ According to the VAT document (Cai Shui [2001] 198 and Cai Shui [2008] 156), the VAT of wind power revenue will be half reimbursed.

⁴ Law of Enterprise Income Tax of the People's Republic of China, Order No. 63 of the President of the People's Republic of China;



	4% ⁵ (Educational surtax)	
	5% ⁶ (City maintenance and construction tax)	
Expected CERs price	10.5Euro/tCO ₂ e	Estimated

Annual Electricity output

The annual electricity output in the FSR is calculated based on the wind resources assessment and professional software that is widely used in Feasibility Study of Wind Farm Projects. The wind resources assessment of the proposed project were assessed based on one year wind resource data measured by the anemometer tower and justified with 29 years (1979-2008)) wind resource data from Pingdu Weather Station. The FSR of the proposed project was evaluated and approved by Shandong Development and Reform Commission, including the annual electricity output assessment. In the FSR, the annual theoretical power generation is 139,159 MWh. Considering the correction of 71.16%, the net annual power generation is 99,026 MWh.

Annual O&M Cost

With the related basic information of the proposed project, the annual O&M cost in FSR is estimated according to relative national standards and regulations. The annual O&M cost consist of maintenance cost, salary and welfare of employee, material cost, insurance cost and miscellaneous cost⁷. The FSR is finished by a qualified FSR developer--- Shandong Industrial Design Institute Co., Ltd. and approved by Shandong Development and Reform Commission. Considering all wind farm projects in Shandong Province which have been registered, the highest percentage of annual O&M cost in the total investment cost is 4.81%⁸, and the proposed project is 2.19%, which is lower than 4.81%. Thus it is demonstrated that the estimated annual O&M cost is reasonably applied in the financial analysis.

Tariff

On 23/07/2008, the first tariff notification (Document Fagaijiage[2008]1876)⁹ regarding to wind power projects in Shandong province was issued by National Development and Reform Commission (NDRC). In this notification, the tariff of 10 wind power projects in Shandong province was approved

⁵ Order No.448 of the State Council, the amendment of “Interim regulations of Educational Surtax” and “ Interim regulations of Local Educational Surtax in Shandong Province,
<http://www.qdedu.gov.cn/qdedu/6dad12c-bed5-462d-9c49-bd07e8082212/29dab101-0fe1-443d-89ca-591172f9aed7.html>

⁶ Interim regulations of City Maintenance and Construction Surtax, Guofa[1985]19.

⁷ Miscellaneous cost is the costs which are additional to the depreciation, amortization, maintenance, insurance, material, welfare and financial costs, including administration costs, travelling expenses, costs of labour-union, staff training expense, cost of board meetings, occupation & health safety costs, business entertainment expense, pollution discharge expenses, technical development expense, land use tax, house tax, vehicle and vessel tax, stamp duty etc.

⁸ Referring to all the PDD of wind farm projects in Shandong Province which have been registered until now, 23 wind farm projects were referenced. The highest percentage of annual O&M cost in the total investment cost is from Qixia Tangshanpeng Wind Farm Project (1019).

⁹ Document issued by NDRC (Fagaijaige[2008]1876),

http://www.ndrc.gov.cn/zcfb/zcfbtz/2008tongzhi/t20080813_230718.htm



to be 0.61 RMB/kWh (incl. VAT).

On 20/07/2009, NDRC released the “Circular on Improving Wind Power On-grid Tariff Policy” (Fagaijiage [2009]1906)¹⁰, which clarified the policy for wind farm tariffs. Four different wind resource regions were defined based on wind resource status and project construction conditions with corresponding guiding tariffs. The proposed project belongs to the forth wind resource region, and the tariff is 0.61 RMB/kWh.

Therefore, the tariff of 0.61Yuan/kWh was used in the FSR (and thus also used in the PDD), which was the latest available on-grid tariff at the time of the preparation of the FSR.

This is the tariff listed in the “Information note on the highest tariffs applied by the Executive Board in its decisions on registration of projects in the People’s Republic of China” (version 02). The highest tariff in Shandong Province is 0.61 Yuan/kWh, therefore, the tariff applied is reasonable and conservative.

(2) Comparison of IRR for the proposed project and the financial benchmark

In accordance with the benchmark analysis (Option III), the proposed project will not be considered as financially attractive if its financial indicators (such as Project IRR) are lower than the benchmark.

Table 1 shows the IRRs of the proposed project, with and without CDM-related income. Without CDM-related income, the Project IRR is lower than the benchmark and the proposed project is not financially acceptable. With it, the Project IRR is better than the benchmark and therefore, the proposed project is financially acceptable.

Table 1. Financial indicators of the Proposed Project

	Project IRR (Total investment, benchmark=8%)
Without CDM-related income	5.63%
With CDM-related income	8.36%

Sub-step 2d. Sensitivity analysis (only applicable to options II and III):

The purpose of the sensitivity analysis is to assess the impact of uncertainties in the input values of the financial model on the calculated IRR. According to “Guideline on the Assessment of Investment Analysis” (version 05), *only variables, including the initial investment cost, that constitute more than 20% of either total project costs or total project revenues should be subjected to reasonable variation. As a general point of departure variations in the sensitivity analysis should at least cover a range of +10% and -10%.*

Because total static investment and annual O&M cost are more than 20% of the total investment. And the tariff and annual power output are selected as the factors for the reason that these two parameters decide the total project revenues. Therefore, four financial parameters including: total static investment, annual O&M cost, tariff and net electricity generation were identified as the main variable factors for sensitive analysis of financial attractiveness. The sensitive analysis will be conducted with the fluctuation of the four parameters with the range from -10% to +10%.

¹⁰ Document issued by NDRC (Fagaijiage [2009]1906),
http://www.ndrc.gov.cn/jgg/jggs/t20090727_292846.htm

Based on the analysis in the FSR, for a wind farm project without CDM funding, the critical factors that influences the Project IRR are mainly as follows:

- 1) Total static investment;
- 2) Annual O&M cost;
- 3) Net power Supply;
- 4) Tariff

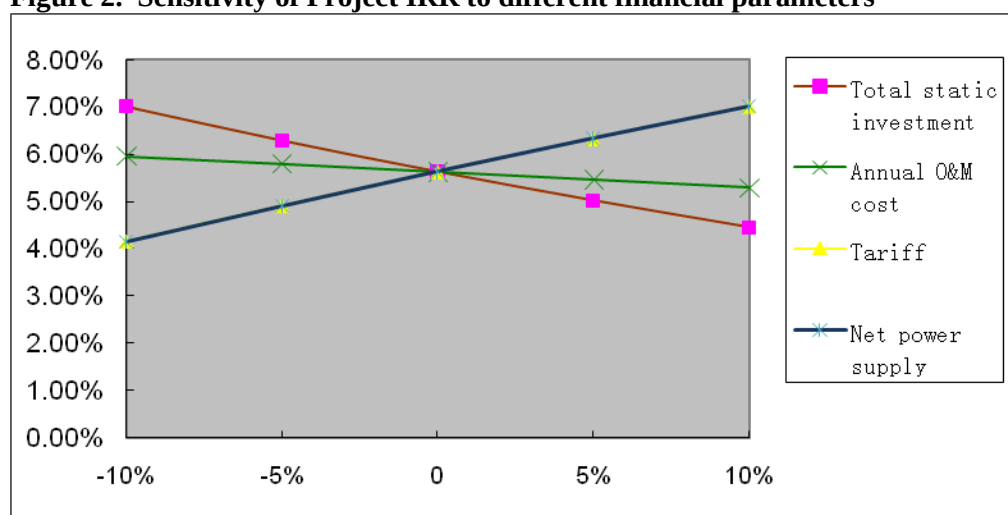
They are fluctuated within the range from -10% to +10% in the FSR and the selection is also in accordance with the Guidance from “Tool for the demonstration and assessment of additionality” (Version 05.2.1). Their impacts on Project IRR of the proposed project were presented in Table 2.

They are fluctuated within the range from -10% to +10% in the FSR and the selection is also in accordance with Guidelines on the Assessment of Investment Analysis” (version 05.0). Their impacts on Project IRR of the proposed project were presented in Table 2 and Figure 2.

Table 2. Sensitivity of Project IRR to different financial parameters

Parameter	Range				
	-10%	-5%	0	5%	10%
Total static investment	7.01%	6.29%	5.63%	5.02%	4.45%
Annual O&M cost	5.96%	5.80%	5.63%	5.46%	5.29%
Net power supply	4.15%	4.91%	5.63%	6.33%	7.02%
Tariff	4.15%	4.91%	5.63%	6.33%	7.02%

Figure 2. Sensitivity of Project IRR to different financial parameters



As shown in Table 2 and Figure 2, the Project IRR of the proposed project varies to different extents, when the above four financial indicators fluctuate within the reasonable range from -10% to +10%. The project IRR does not exceed the benchmark of 8% when the annual O&M cost fluctuates within range from -10% to 10%. The project IRR reaches the benchmark 8% when the total static investment



decreases by 16.2%, the net electricity generation and tariff increase by 17.7%.

Total static investment

When the total static investment decreases by 10%, the Project IRR is 7.01%, lower than benchmark 8%. If the total static investment decreases by 16.2%, the Project IRR reaches the benchmark. Considering the total static investment happened actually of the proposed project during the construction period. The actual cost is 90.11% of the total static investment in the FSR. Therefore, the total static investment of the proposed project is impossible to be decreased by 16.2% to reach the benchmark.

Annual O&M cost

When the annual O&M cost decreases by 10%, the Project IRR is 5.96% lower than the benchmark. If the annual O&M cost decreases by 73.8%, the Project IRR reaches the benchmark. Yearbook 2009 the Producer Price Index for Manufactured Goods from the year 2004 to 2008 was 106.1, 104.9, 103.0, 103.1 and 106.9¹¹ respectively, and also the average salary of labors gradually increased from the year 2004 to 2008¹², which indicates that the operation and maintenance cost is always increasing. Therefore, annual O&M Cost is unlikely to decrease by as much as 73.8%.

Net power supply

When the net power supply increases by 10%, the Project IRR is 7.02%, lower than the benchmark. The figure shows that if there is an increase in power supply by 17.7%, the project IRR begins to exceed the benchmark. However this growth rate is unlikely to occur. Because wind data from FSR shows that wind resources at the project site keep stable¹³. The annual electricity generation in the FSR is estimated based on the long term wind data (from 1979 to 2008) analysis. The project IRR may achieve the benchmark only if the annual electric output increases by 17.7% during the 20 years through the life time of the project, which is impossible to happen.

Tariff

When the tariff increases by 10%, the Project IRR is 7.02%, lower than the benchmark. The figure shows that if there is an increase in tariff by 17.7%, which is as high as 0.718 RMB/kWh, the IRR begins to exceed the benchmark. However, considering the tariff trend from 2002 in Shandong province which the project located, it seems that the tariff is very unlikely to achieve in order to reach the desired benchmark of 8%. The detail analysis of the approved tariffs of all Shandong wind farm projects since the Power Sector Reform in March 2002 are listed in Table 3 below.

Table 3: Approved tariffs of all wind farms in Shandong Province since the Power Sector Reform

¹¹ Source: chapter 9-11, China statistic yearbook 2009

¹² Source: chapter 10-5, China statistic yearbook 2009

<http://www.stats.gov.cn/tjsj/ndsj/2010/indexch.htm>

¹³ Feasibility Study Report, P₂₆



Project	Approved tariff (with VAT) Yuan/kWh	Reference	CDM status / Note
Shandong Tuoji Island Wind Farm Project	0.61	Fa Gai Jia Ge [2008]1876 issued on July 23 2008 by NDRC	Ref No.1789
Shandong Haiyang Qiwershan Wind Power Project	0.61		Ref No.2021
Shandong Weihai 69MW Wind Power Project	0.61		Ref No.1128
Rongcheng 48.75MW Wind Power Project	0.61		Ref No.1755
Shandong Changdao 27.2MW Wind Power Project	0.61		Ref No.1090
Rongcheng Dongchudao Wind Farm	0.61		Ref No.1008
Laizhou Diaolongzui Wind Farm	0.61		Ref No.1010
Qixia Tangshapeng Wind Farm Project	0.61		Ref No.1019
Hudian Laizhou Wind Farm Project	0.61		Ref No.2831
Shandong Laizhou Phase I Wind Power Project	0.61		Ref No.2530
Shandong penglai pingdingshan	0.61	Lu Jia Ge Fa [2009]50 issued on March 13 2009 by Shandong Pricing Bureau	Ref No.2397
Yantai dongyuan laizhou 48.5 MW wind farm			Ref No.2764
Shandong Weihai 69MW Wind Power Project			Ref No.1128
Huaneng Shouguang Wind Power Project			Ref No.3391
Shandong dongying phase 1			Ref No.2584
Huaneng Shandong Shouguang Phase II Wind Power Project			Ref No.5252
Huaneng Shandong Shouguang Phase III Wind Power Project			Ref No.5269
Shandong Liuwangzhuang Wind Farm Project			Ref No.5006
Huaneng Shandong Hekou Phase III Wind Farm Project			Ref No.4830
Huaneng Shandong Rongcheng Chudao Wind Power Project			Ref No.4699
Huaneng Shandong Changyi The proposed project	0.61	Lu Jia Ge Fa [2009]238 issued on December 11 2009 by Shandong Pricing Bureau	Ref No.3353
Yantai dongyuan laizhou 22.5 MW wind farm			under validation



Guohua Dongying Hekou 49.5 MW Wind Farm Project (Phase 1)			Ref No.2436
Anqiu taipingshan 48MW	0.61	Lu Jia Ge Fa [2009]250 issued on December 30 2009 by Shandong Pricing Bureau	NA
Guohua Binzhou Zhanhua 49.5MW Wind Farm Project (Phase 1)			Ref No.2438
Guohua Dongying Lijin 49.5 MW Wind Farm Project (Phase 1)			Ref No.2442
Shandong laizhou phase 2			Ref No.2730
Shandong luneng dongying 48MW wind farm project			Ref No.2019
Datang wendeng Wind Power Project			under validation
Shandong Penglai Wind Farm Phase II Project			Ref No.4402
Malanshan 1.65MW wind farm project	0.61	Lu Jia Ge Fa [2010]32 issued on February 8 2010 by Shandong Pricing Bureau	not CDM
Huaneng Shandong Hekou Phase II Wind Farm Project			Ref No.5081
Huaneng Shandong Hekou Phase III Wind Farm Project			Ref No.4830
Shandong Huaneng Hekou Phase I Wind Farm Project			Ref No.3718
Huaneng shandong binhai wind farm	0.61	Lu Jia Ge Fa [2010]90 issued on April 26 2010 by Shandong Pricing Bureau	under validation

Source: Statistics of installed capacity of wind power in China in 2007, Professor Shi Pengfei (for projects installed by 31 December 2007), the tariff notifications (Fa Gai Jia Ge [2008]1876/ Lujiagefa [2009]50/ Lujiagefa [2009]238/Lujiagefa [2009]250/ Lujiagefa [2010]90/ Lujiagefa [2010]32/)

(<http://www.wwfchina.org/english/downloads/Wind%20Report/china%20wind%20power%202008.pdf>), and UNFCCC web site.

As shown in the table above, all projects since the Power Sector Reform have received the same tariff: 0.61 RMB/kWh. In 2009¹⁴, China's Wind Power Benchmark on Tariff was issued by NDRC, according to which, the region of proposed projects was classified as the forth wind resource zone and the tariff is 0.61 RMB/kWh. Moreover, According to the "Information Note on the Highest Tariffs Applied by the Executive Board in its decisions on Registration of Projects in the People's Republic of China"(version 02), the highest tariff in Shandong Province is 0.61 RMB/kWh. As this result, the tariff in Shandong is deemed to be stable and it is impossible for the proposed project to get a tariff increased by 17.7% and the tariff applied is reasonable and conservative.

¹⁴ http://www.sdpc.gov.cn/zcfb/zcfbtz/2009tz/t20090727_292827.htm



In conclusion of the sensitive analysis, as the financial indicators vary within reasonable range, the proposed project remains financially unacceptable without CDM support and the proposed project is additional.

Step 4. Common practice analysis

Sub-step 4a. Analyze other activities similar to the proposed project activity:

Choose Similar Area

In China, the general environment of projects of wind power projects such as the wind resources, tariff, and investment climate are only similar and comparable in the same province. Therefore, the common practice region and comparable framework is provincial and the proposed project is compared to other projects in Shandong Province.

Choose similar year

In April 2002, China implemented the policy "Separate power plants from network and compete in price to enter network"¹⁵ The objective of this power sector reform is to establish a more commercialized power market in China. Power project investment has to be under a more commercialized condition and considers project investment return more seriously. Since market condition for wind power project development changes much since April 2002, this common practice analysis starts from April 2002.

Choose similar scale

Besides, wind farm within 15MW is defined as small scale CDM project in CDM guidance, which means wind farm project of 15MW or smaller is considered to be very small comparing with big project. Hence, the bottom range for comparison with the proposed project is defined to 15MW.

Table 3 shows the similar projects commissioned since 2002

Table 3 shows the similar projects commissioned since 2002 and have not applied for CDM project.

Table 5 Similar Wind Farm Development in Shandong Province (since 2002)¹⁶

No.	Project	Total capacity
1	Shandong Qingdao Huawei Wind Power Project	16.4MW

Sub-step 4b. Discuss any similar options that are occurring:

As for Table 3, since the power sector reform of 2002, the only large scale project which is not CDM is the Shandong Qingdao Huawei Wind Power Project. This project was an early demonstration project and received ODA support from DEG¹⁷ (part of KfW Bankengruppe). It also faces financial

¹⁵ China implemented the policy "Separate power plants from network and compete in price to enter network"

¹⁶ Reference: *Statistic of installed capacity of China's Wind Farms in 2007, 2008 and 2009* issued by Chinese Wind Energy Association

¹⁷ Source: Page 5 in: <http://www.pd-forum.net/files/0be74ab648713381bb267005dc7e208e.pdf>



barriers and applies for Voluntary Emission Reductions¹⁸.

The other wind power projects all have been applying for CDM support or have been successfully registered as CDM project which are listed in the table below. CDM-related income is an effective measure for the project developers to overcome the barriers they encounter during project development. Hence, the development of this type of project activity in Shandong Province is hard to be brought forth without CDM support.

Ref.	Title	Installed capacity (MW)	Power Supplied (MWh)	Total investment (million yuan)	IRR without CDM
1008	Rongcheng Dongchudao Wind Farm	15.0	28,529	173.41	6.95%
1010	Laizhou Diaolongzui Wind Farm	48.8	96,720	450.14	6.56%
1019	Qixia Tangshanpeng Windfarm Project	21.8	36,580	162.00	7.17%
1090	Shandong Changdao 27.2 MW Wind Power Project	27.2	58,154	268.68	6.59%
1128	Shandong Weihai 69 MW Wind Power Project	69.0	140,553	705.11	6.72%
1755	Roncheng Wind Power Project, 48.75MW	48.8	88,310	502.00	7.06%
1789	Shandong Tuoji Island Windfarm Project	12.0	23,628	101.92	7.31%
2019	Shandong Luneng Dongying 48MW Wind Power Project	48.0	95,136	520.58	6.35%
2021	Shandong Haiyang Qiuershan Wind Power Project	15.0	28,890	154.03	6.63%
2094	The 48MW wind power project in Lubei, Shandong Province	48.0	98,520	539.05	6.62%
2397	Shandong Penglai Pingdingshan Wind Farm Project	48.0	101,904	582.70	5.97%
2436	Guohua Dongying Hekou 49.5 MW Wind Farm Project (Phase 1)	49.5	102,396	507.58	5.91%
2438	Guohua Binzhou Zhanhua 49.5MW Wind Farm Project (Phase 1)	49.5	99,324	479.77	5.98%
2442	Guohua Dongying Lijin 49.5 MW Wind Farm Project (Phase 1)	49.5	103,140	510.32	5.95%
2530	Shandong Laizhou phase I Wind Power Project	49.5	96,080	516.30	6.36%
2584	Shandong Dongying 1st phase Wind Power Project	49.5	95,978	639.24	6.90%
2764	Yantai Dongyuan Laizhou 48.5	48.5	88,187	449.24	6.64%

¹⁸The verification report of Shandong Qingdao Huawei Wind Power Project, Report No.20070225.
<http://www.bestonecapital.com/chinese/03.htm>



	MW Wind Farm Project Phase I				
2814	Shandong Rushan Luneng Wind Farm	42.0	69,125	389.39	5.97%
2831	Huadian Laizhou Wind Farm Project	40.5	77,320	425.56	5.58%
3353	Huaneng Changyi Phase I Wind Farm Project	49.5	111,875	624.48	5.30%
3391	Shandong Huaneng Shouguang 49.5MW Wind Farm Project	49.5	96,479	559.07	6.97%
3603	Huaneng Binhai Wind Farm Project	49.5	114,114	630.35	5.54%
3616	Yantai Dongyuan 25.5 MW Wind Farm Project Phase I	25.5	51,689	251.95	6.04%
3718	Shandong Huaneng Hekou Phase I Wind Farm Project	49.5	100,090	545.31	5.00%
4402	Shandong Penglai Wind Farm Phase II Project	48.0	103,740	539.62	6.72%
4587	Penglai Daliuhang Wind Farm Project Phase I	49.8	103,883	550.55	4.96%
4612	Weihai Jingqu 49.8 MW Wind Farm Project Phase I	49.8	102,986	541.95	5.07%
4614	Huaneng Laizhou Phase II Wind Farm Project	49.5	89,100	509.29	3.64%
4699	Huaneng Shandong Rongcheng Chudao Wind Power Project	48.0	90,240	492.61	4.62%
4830	Huaneng Shandong Hekou Phase III Wind Farm Project	49.5	100,023	549.00	4.48%
4871	Huaneng Shandong Hekou Phase IV Wind Farm Project	49.5	99,488	545.81	4.50%
4882	Guohua Rongcheng Phase II Wind Farm Project	49.5	102,337	535.95	6.20%
4923	Shandong Binzhou Zhanhua Tao'erhe Wind Power Project	49.5	96,525	512.13	5.57%
5006	Shandong Liuwangzhuang Wind Farm Project	38.25	71,750	390.51	4.62%
5081	Huaneng Shandong Hekou Phase II Wind Farm Project	49.5	101,351	557.51	4.44%
5252	Huaneng Shandong Shouguang Phase II Wind Power Project	49.5	96,030	562.757	4.75%
5269	Huaneng Shandong Shouguang Phase III Wind Power Project	49.5	96,676	563.39	4.82%

As stated above, it is concluded that the proposed project is not common practice within the region. Therefore, the proposed project is additional.

In conclusion, the proposed project is additional and the CDM revenue plays the key role to make the proposed project financially feasible. The CDM was seriously considered in the decision to



implement the project activity.

B.6. Emission reductions:

B.6.1. Explanation of methodological choices:

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The key methodological steps are as follows:

1. Calculating the Baseline Emission (BE_y)
2. Calculating the Project Emission (PE_y)
3. Calculating the Leakage Emission (LE_y)
4. Calculating the Emission Reduction (ER_y)

1. Calculating the Baseline emissions

The baseline emissions (BE_y) is the product of the baseline emissions factor ($EF_{grid,CM,y}$ in tCO₂e/MWh) calculated, times the electricity supplied by the project activity to the grid (EG_y in MWh), as follows:

$$BE_y = EG_{PJ,y} \times EF_{grid,CM,y} \quad (1)$$

1.1 Calculation of the baseline emissions factor

Following ACM0002 (version 12.2.0), the baseline emission factor (EF_y) is calculated as a combined margin ($EF_{grid,CM}$), consisting of the combination of operating margin ($EF_{grid,OM}$) and build margin ($EF_{grid,BM}$) factors according to the following six steps defined in the “Tool to calculate the emission factor for an electricity system” (version 02.2.0). Data for the calculations are based on official national statistics books: China Energy Statistical Yearbook and China Electric Power Yearbook.

Step 1. Identify the relevant electricity systems

The DNA of the host country has published a delineation of the project electricity system and connected electricity systems, this delineation is used. Following the DNA delineation, the project electricity system is the North China Power Grid (NCPG), which consists of Beijing, Tianjin, Hebei, Shanxi, Inner Mongolia Autonomous Region, and Shandong Province Power Grids. The proposed project is located in Shandong Province and covered by the NCPG. Therefore, NCPG is chosen as the relevant electric power system.

The connected electricity system is Central China Power Grid, which is connected by transmission lines to the project electricity system. Power plants within the connected electricity system can be dispatched without significant transmission constraints but transmission to the project electricity system has significant transmission constraint.

Electricity transfers from connected electricity system to the project electricity system are defined as electricity imports and electricity transfers to connected electricity systems are defined as electricity exports. The NCPG has the electricity imports from the Central China Power Grid.

For the purpose of determining the build margin emission factor, the spatial extent is limited to the project electricity system (NCPG), since the electricity imports from Central China Power Grid account for a very small percentage and rent or likely future additions to transmission capacity will



not enable significant increases in imported electricity.

For the purpose of determining the operating margin emission factor, one of the following options could be used to determine the CO₂ emission factor(s) for net electricity imports ($EF_{grid, import, y}$) from a connected electricity system within the same host country:

- (a) 0 tCO₂/MWh, or
- (b) The weighted average operating margin (OM) emission rate of the exporting grid, determined as described in step 4 below; or
- (c) The simple operating margin emission rate of the exporting grid, determined as described in step 4 below; or
- (d) The simple adjusted operating margin emission rate of the exporting grid, determined as described in step 4 below.

For the proposed project activity, option (b) is used to determine the CO₂ emission factor(s) for net electricity imports ($EF_{grid, import, y}$) from Central China Power Grid, since the conditions for this method described in step 3 below are applied to the exporting grid.

STEP 2. Choose whether to include off-grid power plants in the project electricity system (optional)

According to the Approval of electricity connection to North China Power Grid, all the power generated by the project activity will be supplied to the power grid company. Thus, the proposed project does not include off-grid power plants in the project electricity system referred in apply to “Tool to calculate the emission factor for an electricity system”.

Step 3. Select a method to determine the operating margin (OM)

The calculation of the operating margin emission factor ($EF_{grid, OM, y}$) is based on the following methods:

- (a) Simple OM, or
- (b) Simple adjusted OM, or
- (c) Dispatch Data Analysis OM, or
- (d) Average OM.

Detailed information to carry out a dispatch data analysis is not publicly available; therefore, method (b) and method (c) is not suitable for the proposed project.

According to ACM0002, the Simple OM method is applicable to the project if the low-cost resources constitute less than 50% of total grid generation on average in the five most recent years or based on long-term normals for hydroelectric production.

The North China Power Grid only possesses 1.188% of its total electricity generation that comes from renewable energy sources in 2008, 0.752% in 2007, 0.745% in 2006, 0.749% in 2005 and 0.708% in 2004¹⁹. Hence, the low operating cost/must run sources is much less than 50% of the total grid generation, Consequently, Simple OM method is selected to calculate the Operating Margin emission

¹⁹ China Electric Power Yearbook 2005-2009



factor (OM) of the proposed project.

The Simple OM can be calculated using either of the two following data vintages for years(s) y:

- (Ex-ante option): If the ex ante option is chosen, the emission factor is determined once at the validation stage, thus no monitoring and recalculation of the emissions factor during the crediting period is required. For grid power plants, use a 3-year generation-weighted average, based on the most recent data available at the time of submission of the CDM-PDD to the DOE for validation.
- (Ex-post option): If the ex post option is chosen, the emission factor is determined for the year in which the project activity displaces grid electricity, requiring the emission factor to be updated annually during monitoring.

Here ex-ante vintage is chosen, and the $EF_{grid,OM}$ is fixed during the first crediting period.

Step 4. Calculate the operating margin emission factor according to the selected method

The Simple OM emission factor is calculated as the generation-weighted average CO₂ emissions per unit net electricity generation (tCO₂/MWh) of all generating power plants serving the system, not including low-cost / must-run power plants / units. The simple OM may be calculated:

- Option A: Based on the net electricity generation and a CO₂ emission factor of each power unit, or
- Option B: Based on the total net electricity generation of all power plants serving the system and the fuel types and total fuel consumption of the project electricity system.

For the proposed project activity, the required data for the exercise of Option A is not available and those of Option B can be obtained from official sources, and off-grid power plants are not included in the calculation, therefore, Option B is chosen to calculate the operating margin emission factor:

For Option B, the Simple OM emission factor is calculated based on the net electricity supplied to the grid by all power plants serving the system, not including low-cost / must-run power plants / units, and based on the fuel type(s) and total fuel consumption of the project electricity system, as follows:

$$EF_{grid,OMsimple,y} = \frac{\sum_{i,m} FC_{i,y} \cdot NCV_{i,y} \cdot EF_{CO2,i,y}}{\sum_m EG_y} \quad (2)$$

Where:

$EF_{grid,OMsimple,y}$	=	Simple operating margin CO ₂ emission factor in year y (tCO ₂ e/MWh)
$FC_{i,y}$	=	Amount of fossil fuel type <i>i</i> consumed in the project electricity system in year y (mass or volume unit)
$NCV_{i,y}$	=	Net calorific value (energy content) of fossil fuel type <i>i</i> in year y (GJ / mass or volume unit)
$EF_{CO2,i,y}$	=	CO ₂ emission factor of fossil fuel type <i>i</i> in year y (tCO ₂ e/GJ)
EG_y	=	Net electricity generated and delivered to the grid by all power sources serving the system, not including low-cost / must-run power plants / units, in



		year y (MWh)
i	=	All fossil fuel types combusted in power sources in the project electricity system in year y
y	=	The relevant year as per the data vintage chosen in Step 3

For this approach (simple OM) to calculate the operating margin, the simple OM emission factor is calculated based on the net electricity supplied to the grid by all power plants serving the system, not including low-cost / must-run power plants / units, and including electricity imports to the grid. Electricity imports should be treated as one power plant source.

Regarding parameter selection, local values of $NCV_{i,y}$ and $EF_{CO_2,i,y}$ should be used where available. If no such values are available, IPCC world-wide default values are preferable. The Net Calorific Value ($NCV_{i,y}$) of each type of fossil fuel used in the calculation comes from China Energy Statistic Yearbook 2009. Emission factors ($EF_{CO_2,i,y}$) of each type of fossil fuel come from IPCC 2006 default values.

On the basis of the data available, the three-year average operating margin emission factor is calculated by the DNA as a full-generation-weighted average of the emission factors:

$$EF_{grid,OMsimple} = 0.9914 \text{ tCO}_2\text{e/MWh}$$

Step 5. Calculate the build margin emission factor

In terms of vintage of data, project participants can choose between one of the following two options:

Option 1: For the first crediting period, calculate the build margin emission factor *ex ante* based on the most recent information available on units already built for sample group m at the time of CDM-PDD submission to the DOE for validation. For the second crediting period, the build margin emission factor should be updated based on the most recent information available on units already built at the time of submission of the request for renewal of the crediting period to the DOE. For the third crediting period, the build margin emission factor calculated for the second crediting period should be used. This option does not require monitoring the emission factor during the crediting period.

Option 2: For the first crediting period, the build margin emission factor shall be updated annually, *ex post*, including those units built up to the year of registration of the project activity or, if information up to the year of registration is not yet available, including those units built up to the latest year for which information is available. For the second crediting period, the build margin emissions factor shall be calculated *ex ante*, as described in Option 1 above. For the third crediting period, the build margin emission factor calculated for the second crediting period should be used.

In this PDD, Option1 is chosen to calculate the build margin emission factor.

The build margin emission factor is the generation-weighted average emission factor (tCO_2/MWh) of all power units m during the most recent year y for which power generation data is available, calculated as follows:

$$EF_{grid,BM,y} = \frac{\sum_m EG_{m,y} \times EF_{EL,m,y}}{\sum_m EG_{m,y}} \quad (3)$$

Where:

$EF_{grid,BM,y}$	=	Build margin CO ₂ emission factor in year y (tCO ₂ e/MWh)
$EG_{m,y}$	=	Net quantity of electricity generated and delivered to the grid by power unit m in year y (MWh)
$EF_{EL,m,y}$	=	CO ₂ emission factor of power unit m in year y (tCO ₂ e/MWh)
m	=	Power units included in the build margin
y	=	Most recent historical year for which power generation data is available

Currently, it is very difficult to get the capacity margin data of power plants in China, since these data as well as net quantity of electricity generated and delivered to the grid and fuel consumption data in power unit m are regarded as commercial secrets or only for internal usage. Then the following deviation²⁰ approved by the EB was adopted to calculate the Build Margin emission factor.

According to the guidance from the CDM Executive Board for a deviation of the baseline methodology of AM0005, which had combined into the baseline methodology of ACM0002, the following deviation was adopted to calculate the Build Margin emission factor.

- 1) Use the efficiency level of the best technologies commercially available in the provincial/regional or national grid of China, as a conservative proxy, for fuel i consumption estimation to estimate the $EF_{grid,BM,y}$.
- 2) Use capacity additions during last several years for estimating the $EF_{grid,BM,y}$, i.e. the capacity addition over last several years, whichever results in a capacity addition that is closest to 20% of total installed capacity. For the proposed project, the data from Year 2006 to 2008 is used to calculate $EF_{grid,BM,y}$.
- 3) Use installed capacity to replace annual power generation to estimate weights.

The BM emission factor in this PDD is calculated as following sub-steps.

Sub-step 1. Calculation of weights of CO₂ emissions of solid, liquid and gaseous fossil fuels in total emissions for power generation

$$\lambda_{Coal,y} = \frac{\sum_{i \in COAL,j} F_{i,j,y} \times NCV_{i,y} \times EF_{CO_2,i,y}}{\sum_{i,j} F_{i,j,y} \times NCV_{i,y} \times EF_{CO_2,i,y}} \quad (4)$$

²⁰Source: http://cdm.unfccc.int/UserManagement/FileStorage/AM_CLAR_QEJWJEF3CFBP1OZAK6V5YXPQ_KK7WYJ

$$\lambda_{Oil,y} = \frac{\sum_{i \in OIL,j} F_{i,j,y} \times NCV_{i,y} \times EF_{CO_2,i,y}}{\sum_{i,j} F_{i,j,y} \times NCV_{i,y} \times EF_{CO_2,i,y}} \quad (5)$$

$$\lambda_{Gas,y} = \frac{\sum_{i \in GAS,j} F_{i,j,y} \times NCV_{i,y} \times EF_{CO_2,i,y}}{\sum_{i,j} F_{i,j,y} \times NCV_{i,y} \times EF_{CO_2,i,y}} \quad (6)$$

Where:

$FC_{i,j,y}$ = Amount of fossil fuel type i consumed in province j in year y
(mass or volume unit)

$NCV_{i,y}$ = Net calorific value (energy content) of fossil fuel type i in year y (GJ/t or GJ/m³)

$EF_{CO_2,i,y}$ = CO₂ emission factor of fossil fuel type i in year y (tCO₂e/GJ)

Coal, *Oil* and *Gas* refer to the group of solid, liquid, and gaseous fossil fuels, respectively.

Sub-step 2: Calculation of Emission Factor of Relevant Thermal Power

$$EF_{Thermal,y} = \lambda_{Coal,y} \times EF_{Coal,Adv,y} + \lambda_{Oil,y} \times EF_{Oil,Adv,y} + \lambda_{Gas,y} \times EF_{Gas,Adv,y} \quad (7)$$

Where:

$EF_{Coal,Adv}$, $EF_{Oil,Adv}$ and $EF_{Gas,Adv}$ refer to the emission factors representing best technologies commercially available for coal, oil and gas fired power plants, respectively.

Sub-step 3: Calculation of BM of the Grid

Using the share of different type of capacity in total capacity addition as weight, the weighted average of emission factors of different type capacity is calculated as the Build Margin emission factor $EF_{grid,BM,y}$ of North China Power Grid.

$$EF_{grid,BM,y} = \frac{CAP_{Thermal,y}}{CAP_{Total,y}} \times EF_{Thermal,y} \quad (8)$$

Where:

CAP_{Total} = The total newly added electricity generation capacity (MW)

$CAP_{Thermal}$ = The newly added electricity generation capacity of thermal power (MW)

Following the four steps above, the build margin emission factor $EF_{grid,BM,y}$ of the NCPG is calculated to be: 0.7495 tCO₂e/MWh (see Annex 3 for more details).

Step 6. Calculate the combined margin emission factor



The baseline emissions factor (EF_{CM}) is calculated as the weighted average of the Operating Margin emission factor and Build Margin emission factors following ACM0002. For wind projects, the default weights are as follows: $w_{OM} = 0.75$ and $w_{BM} = 0.25$:

$$EF_{grid,CM,y} = EF_{grid,OM,y} \times w_{OM} + EF_{grid,BM,y} \times w_{BM} \quad (9)$$

Where:

$EF_{grid,BM,y}$	=	Build margin CO ₂ emission for the project electricity system factor in year y (tCO ₂ e/MWh)
$EF_{grid,OM,y}$	=	Operating margin CO ₂ emission factor for the project electricity system in year y (tCO ₂ e/MWh)
w_{OM}	=	Weighting of operating margin emissions factor (%)
w_{BM}	=	Weighting of build margin emissions factor (%)

On the basis of these weights for the first crediting period, the combined margin emission factor is calculated, and fixed ex-ante:

$$EF_{grid,CM,y} = 0.9914 \times 0.75 + 0.7495 \times 0.25 = 0.9309 \text{ tCO}_2/\text{MWh}$$

Baseline emissions (BE_y) now can be calculated as the combined margin CO₂ emission factor ($EF_{grid,CM,y}$) multiplied by the annual net generation of the Proposed Project ($EG_{PJ,y}$).

$$BE_y = EG_{PJ,y} \times EF_{grid,CM,y}$$

Where:

BE_y = Baseline emissions in year y (tCO₂/yr)

$EG_{PJ,y}$ = Quantity of net electricity generation that is produced and fed into the grid as a result of the implementation of the CDM project activity in year y (MWh/yr)

$EF_{grid,CM,y}$ = Combined margin CO₂ emission factor for the project electricity system in year y calculated using the latest version of the “Tool to calculate the emission factor for an electricity system”.

Calculation of $EG_{PJ,y}$

The proposed project is the installation of a new grid-connected renewable power plant/unit at a site where no renewable power plant was operated prior to the implementation of the project activity, then:

$$EG_{PJ,y} = EG_{facility,y} \quad (10)$$

where:

$EG_{PJ,y}$	=	Quantity of net electricity generation that is produced and fed into the grid as a result of the implementation of the CDM project activity in year y (MWh/yr)
$EG_{facility,y}$	=	Quantity of net electricity generation supplied by the project plant/unit to the grid in year y (MWh/yr)

2 Calculating the Project Emission (PE_y)

According to ACM0002 (version 12.2.0), project emission of the proposed wind farm project is zero. $PE_y = 0$.

3 Calculating the Leakage Emission (LE_y)



According to ACM0002 (version 12.2.0), no leakage is considered. The main emissions potentially giving rise to leakage are neglected.

4 Calculating the Emission Reduction (ER_y)

The annual emission reductions ER_y for the project activity are calculated as the baseline emissions minus the project emissions and minus the leakage emissions. Being the project of a zero-emission activity the final GHG emission reductions are calculated as follows:

$$ER_y = BE_y - PE_y \quad (11)$$

where:

ER_y	=	Emission reductions in year y (t CO ₂ e/yr)
BE_y	=	Baseline emissions in year y (t CO ₂ /yr)
PE_y	=	Project emissions in year y (t CO ₂ e/yr)

B.6.2. Data and parameters that are available at validation:

>>

Data / Parameter:	$FC_{i,m,y}$
Data unit:	tonnes or m ³
Description:	Amount of fossil fuel type i consumed by power plant / unit m in year y
Source of data used:	China Energy Statistical Yearbook (2007~2009)
Value applied:	See Annex 3 for details
Justification of the choice of data or description of measurement methods and procedures actually applied :	Official statistical data
Any comment:	

Data / Parameter:	$NCV_{i,y}$
Data unit:	kJ/kg or kJ/m ³
Description:	Net calorific value (energy content) of fossil fuel type i in year y
Source of data used:	China Energy Statistical Yearbook (2009)
Value applied:	See Annex 3 for details
Justification of the choice of data or description of measurement methods and procedures actually applied :	Official statistical data
Any comment:	

Data / Parameter:	$EF_{CO_2,i,y}$
Data unit:	tc/TJ
Description:	CO ₂ emission factor of fossil fuel type i in year y
Source of data used:	2006 IPCC Guidelines for National Greenhouse Gas Inventories



Value applied:	See Annex 3 for details
Justification of the choice of data or description of measurement methods and procedures actually applied :	IPCC default value
Any comment:	

Data / Parameter:	Installed Capacity
Data unit:	MW
Description:	The Installed Capacity of the power plants in the grid in the year y
Source of data used:	China Electric Power Yearbook (2007-2009)
Value applied:	See Annex 3 for details
Justification of the choice of data or description of measurement methods and procedures actually applied :	Official statistical data
Any comment:	

Data / Parameter:	GEN_y
Data unit:	MWh
Description:	The electricity generation of the power plants in the grid in the year y
Source of data used:	China Electric Power Yearbook (2007-2009)
Value applied:	See Annex 3 for details
Justification of the choice of data or description of measurement methods and procedures actually applied :	Official statistical data
Any comment:	

Data / Parameter:	Efficiency of the best technology commercially
Data unit:	%
Description:	Best commercial available efficiency of coal, gas, oil fuel power plant
Source of data used:	Chinese DNA's Guideline of emission factors of Chinese grids
Value applied:	Best efficiency for coal plant is 39.08% Best efficiency for oil plant is 51.46% Best efficiency for gas plant is 51.46%
Justification of the choice of data or description of measurement methods and procedures actually applied :	Official statistic value



applied :	
Any comment:	Applied for calculating BM

B.6.3 Ex-ante calculation of emission reductions:

>>

As described in B.3., the emission reductions of the proposed project are calculated as follows:

Baseline emissions

Annual generation (net of auxiliary power i.e. the on-site electricity usage for the operation of the wind farm) is estimated as 99,026MWh.

$$EF_{\text{grid,OM,y}} = 0.9914 \text{ tCO}_2/\text{MWh}$$

$$EF_{\text{grid,BM,y}} = 0.7495 \text{ tCO}_2/\text{MWh}$$

$$EF_{\text{grid,CM,y}} = 0.9914 \times 0.75 + 0.7495 \times 0.25 = 0.9309 \text{ tCO}_2/\text{MWh}$$

$$BE_y = 99,026 \times 0.9309 = 92,183 \text{ tCO}_2\text{e}$$

The ex-ante baseline emission factor: **0.9309 tCO₂/ MWh**

Annual baseline emissions: **92,183 tCO₂e** (details in Annex 3)

Project emissions

According to ACM0002 (version 12.2.0), for most renewable power generation project activities, $PE_y = 0$.

Leakage

According to ACM0002 (version 12.2.0), no leakage is considered. The main emissions potentially giving rise to leakage are neglected.

Project Emission Reductions

$$ER_y = BE_y - PE_y$$

The total annual baseline emissions are 92,183 tCO₂.

The total annual project emissions are 0 tCO₂.

$$ER_y = BE_y - PE_y = 92,183 - 0 = 92,183 \text{ tCO}_2$$

The annual emission reductions are estimated to be: 92,183 tCO₂. The proposed project activity is expected to achieve 645,281 tCO₂e of net emission reductions during the first 7-year crediting period.

B.6.4 Summary of the ex-ante estimation of emission reductions:

>>

Year	Estimation of project activity	Estimation of baseline emissions	Estimation of overall emission
------	--------------------------------	----------------------------------	--------------------------------



	emissions (tonnes of CO₂e)	(tonnes of CO₂e)	reductions (tonnes of CO₂e)
20/07/2012-19/07/2013	0	92,183	92,183
20/07/2013-19/07/2014	0	92,183	92,183
20/07/2014-19/07/2015	0	92,183	92,183
20/07/2015-19/07/2016	0	92,183	92,183
20/07/2016-19/07/2017	0	92,183	92,183
20/07/2017-19/07/2018	0	92,183	92,183
20/07/2018-19/07/2019		92,183	92,183
Total (tonnes of CO₂e)	0	645,281	645,281

B.7 Application of the monitoring methodology and description of the monitoring plan:**B.7.1. Data and parameters monitored:**

>>

Data / Parameter:	$EG_{output,y}$
Data unit:	MWh/yr
Description:	Total electricity supplied to the Grid by the proposed project and Phase II project in the year y.
Source of data to be used:	Electricity meter reading in the Tangtian substation
Value of data applied for the purpose of calculating expected emission reductions in section B.5	99,026
Description of measurement methods and procedures to be applied:	The readings of the electricity meter will be continuously measured and monthly recorded. Data will be archived for 2 years following the end of the last crediting period by means of electronic and paper backup. The accuracy of electricity meter will no lower than 0.5s. The meter would be calibrated annually according to relative national standards.
QA/QC procedures to be applied:	The electricity generation from the plant will be monitored and recorded at the central control room. The Grid Company is responsible for recording such data. Receipts for electricity sales will be kept for further verification, when necessary.
Any comment:	

Data / Parameter:	$EG_{another,y}$
Data unit:	MWh/yr
Description:	Electricity generated by the Phase II Project in the year y.
Source of data to be used:	Electricity meter reading on the site of the Phase II project.
Value of data applied for the purpose of calculating expected emission reductions in	0



section B.5	
Description of measurement methods and procedures to be applied:	The readings of the electricity meter will be continuously measured and monthly recorded. Data will be archived for 2 years following the end of the last crediting period by means of electronic and paper backup. The accuracy of electricity meter will no lower than 0.5s. The meter would be calibrated annually according to relative national standards.
QA/QC procedures to be applied:	The electricity generation from the plant will be monitored and recorded at the central control room. The project owner is responsible for recording such data. Receipts for electricity sales will be kept for further verification, when necessary.
Any comment:	

Data / Parameter:	$EG_{import,y}$
Data unit:	MWh/yr
Description:	Total electricity imported from the Grid by the proposed project and Phase II project through the 110kV line in year y.
Source of data to be used:	Electricity meter reading in the Tangtian substation
Value of data applied for the purpose of calculating expected emission reductions in section B.5	0
Description of measurement methods and procedures to be applied:	The readings of the electricity meter will be continuously measured and monthly recorded. Data will be archived for 2 years following the end of the last crediting period by means of electronic and paper backup. The accuracy of electricity meter will no lower than 0.5s. The meter would be calibrated annually according to relative national standards
QA/QC procedures to be applied:	The electricity generation from the plant will be monitored and recorded at the central control room. The Grid Company is responsible for recording such data. Receipts for electricity sales will be kept for further verification, when necessary.
Any comment:	

Data / Parameter:	$EG_{im-spare,y}$
Data unit:	MWh/yr
Description:	Electricity imported from the Grid by the proposed project and Phase II project through a spare 10 kV line in the year y
Source of data to be used:	Derived based on the amount on the sales receipts from local power distributor
Value of data applied for the purpose of calculating expected emission reductions in section B.5	0
Description of measurement methods	The local power distributor will measure the $EG_{im-spare,y}$ by metering instrument operated by the Grid Company. The meter would be calibrated



and procedures to be applied:	per three years according to relative national standards. Electricity sales receipts will be issued accordingly in an approach which is agreed by both local power distributor and the project owner.
QA/QC procedures to be applied:	Reading recording, operation and maintenance of this meter was implemented by the project owner.
Any comment:	

Data / Parameter:	$EG_{facility,y}$
Data unit:	MWh/yr
Description:	Net electricity supplied to the Grid by the proposed project in the year y .
Source of data to be used:	Electricity meter reading at the connection point between the proposed project and the Grid
Value of data applied for the purpose of calculating expected emission reductions in section B.5	99,026
Description of measurement methods and procedures to be applied:	The net electricity supplied to the Grid by the proposed project will be calculated by the following formula: $EG_{facility,y} = EG_{output,y} - EG_{another,y} - EG_{import,y} - EG_{im-spare,y}$
QA/QC procedures to be applied:	-
Any comment:	

B.7.2 Description of the monitoring plan:

>>

This Monitoring plan will set out a number of monitoring tasks in order to ensure that all aspects of projected greenhouse gas (GHG) emission reductions for the proposed project are controlled and reported. This requires an ongoing monitoring of the project to ensure performance according to its design and that claimed Certified Emission Reductions (CERs) are actually achieved.

The proposed project adopts the approved consolidated baseline and monitoring methodology ACM0002 “Consolidated methodology for grid-connected electricity generation from renewable sources” (version 12.2.0) to determine the emission reductions from the net electricity generation from the wind farm. This plan describes in more detail the process.

1. Monitoring Object

The monitoring is to justify the realistic amount of emission reduction from the CDM project. The monitoring plan will provide credible, accurate, transparent and conservative monitoring data and ensure the real, measurable, long-term GHG emission reduction from this project.

2. Management Structure

China Resources Wind Power (Qingdao) Co., Ltd., the project owner will use this document as guideline in monitoring of the project emission reduction performance and will adhere to the guidelines set out in this monitoring plan. This plan should be modified according to actual conditions



and requirements of DOE in order to ensure that the monitoring is credible, transparent, and conservative.

The responsibilities of the project staff are as follow:

The General Manager of the project entity will appoint a CDM project manager. The Operational and Monitoring Manager of the plant, the Financial Chief, and the Technical Chief are responsible for the collection of the data and information required in the monitoring plan.

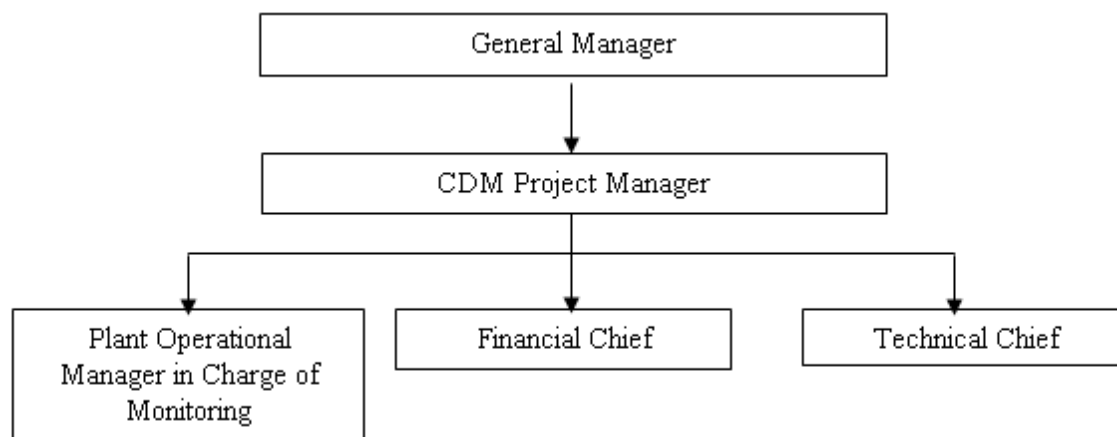
General Manager: To be responsible for supervising the whole monitoring procedure.

CDM Project Manager: To be responsible for data management and compiling monitoring report.

Operational and monitoring manager: To be responsible for collecting data and do internal audit.

Financial chief: To be responsible for collection of sales receipts.

Technical chief: To be responsible for preparing operational reports of the project activity, recording the daily operation of the wind farm, including operating periods, equipment defects, etc.



3. Monitoring Program and Equipments

The electricity supplied by the proposed project to the Grid will be transmitted from the on-site substation of the proposed project through the 110kV transmission line and then be delivered to the Grid and share the 110kV transmission line with the Phase II project, which is developed by the same project owner with the proposed project.

Meter M1 installed on the Tangtian Substation as the main meter will be used to measure the electricity supplied by the proposed project and Phase II project. Meter M2 and M3 installed at the on-site substation will be used to measure the electricity supplied by the proposed project and Phase II project respectively. Sales receipts on the electricity supplied to the grid by Phase II Project will be issued according to meter reading from M3. Sales receipts on the electricity supplied to the grid by the project activity will be issued according to meter reading from M1 minus meter reading from M3. The reading of meter (M2) is carried out by the project owner for cross check. The precision of the meter M1, M2 and M3 will be not lower than 0.5s.

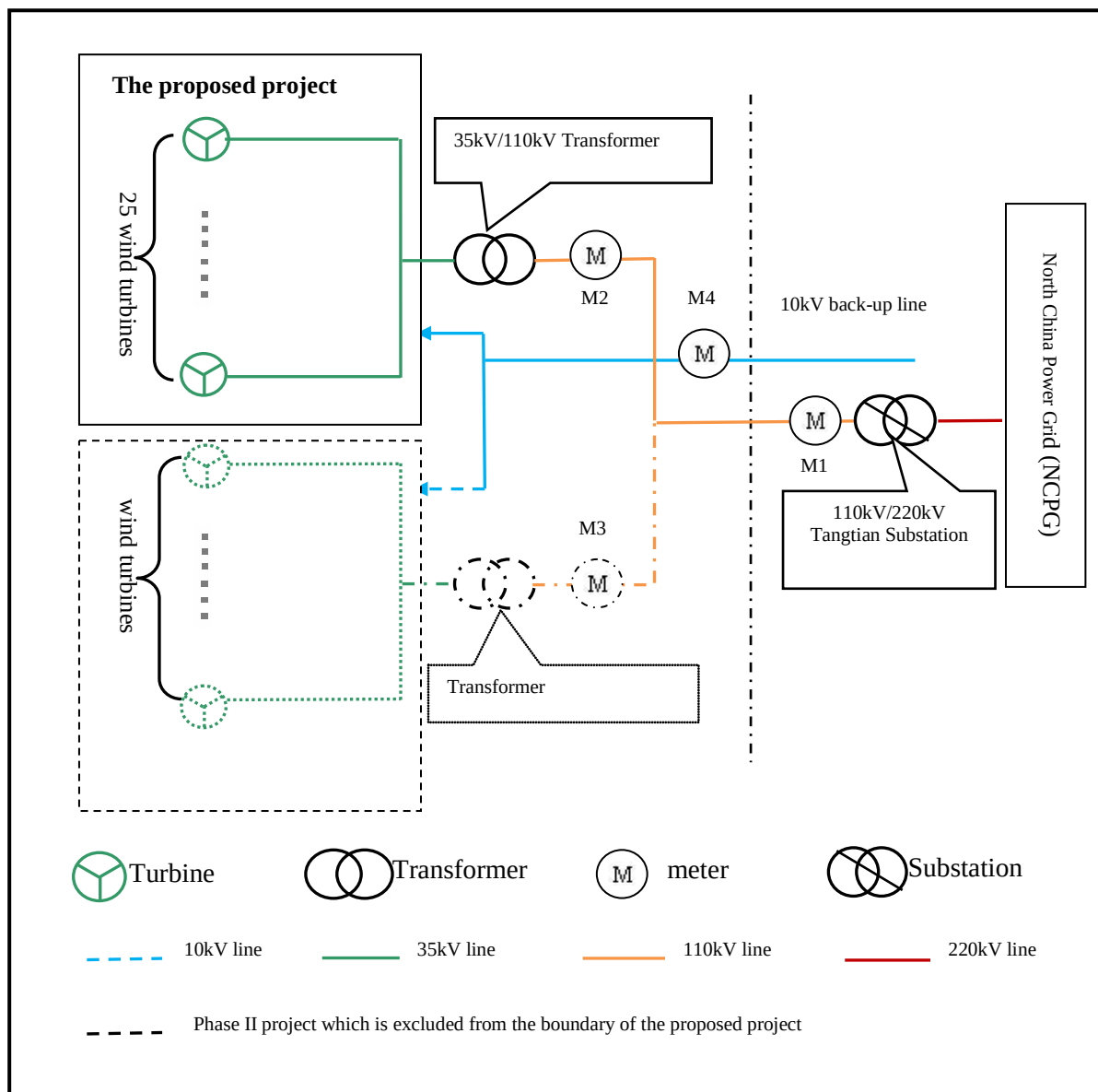
Additionally, a spare 10 kV line is used for supplying electricity in order to start-up or emergencies. Electricity imported from the grid by the project via this spare 10kV line can be measured by a meter



(M4) which is owned by the project owner. Reading recording, operation and maintenance of this meter is implemented by the grid. After electricity supplied to the project by the grid through this spare 10 kV line has been confirmed by both sides (the local power distributor and the project owner), electricity sales receipts will be issued accordingly in an approach agreed from both sides and based on the meter reading. The precision of this meter is no lower than 2.0.

The Grid Company will be responsible for the operation of meter M1; the Project Owner will be responsible for the meter M2 and M3. The metering equipments will be annually calibrated and checked for accuracy by the third party designated according to relevant national calibration standard. An agreement will be signed between the proposed project owner and the Grid that defines the metering arrangements and the required quality control procedures to ensure accuracy. The calibration records will be provided to the proposed project owner and maintained by the proposed project owner and the third party designated.

The location of the meters are as the following chart.



The net electricity supplied to the grid by the proposed project is calculated as followed:

$$EG_{facility,y} = EG_{output,y} - EG_{another,y} - EG_{import,y} - EG_{im-spare,y} \quad (12)$$

Where:

$EG_{facility,y}$: Net electricity supplied to the Grid by the proposed project in the year y .

$EG_{output,y}$: Total electricity supplied to the Grid by the proposed project and Phase II project in the year y, which will be measured by bidirectional meter M1 installed in the Tangtian substation;

$EG_{import,y}$: Total electricity imported from the Grid by the proposed project and Phase II project in the



year y , which will be measured by bidirectional meter M1 installed in Tangtian substation;
 $EG_{im-spare,y}$: Electricity imported from the Grid through a spare 10 kV line in the year y , which will be measured by meter M4 installed on the spare power line on the site of the proposed project
 $EG_{another,y}$: Electricity supplied to the Grid by the Phase II Project in the year y , which will be measured by electricity meter M3;

$EG_{import,y}$ is the total electricity imported from the Grid by the proposed project and the Phase II project in year y , to be conservative, when the $EG_{facility,y}$ is calculated, $EG_{import,y}$ is fully deducted including the electricity imported from the Grid by Phase II project.

In conclusion, the determination of $EG_{facility,y}$ is based on the measurement of electricity meters installed and is conservative.

Before the proposed project is in operation, both the project owner and the grid company will check the equipments to ensure their work properly.

4. Monitoring Training

Monitoring training is critical to ensure that all members of the CDM Monitoring Team has a thorough understanding of the monitoring procedure and are able to carry out the monitoring tasks strictly in line with the CDM requirements. The team leader is responsible for evaluating training outcome. Only qualified staffs can work on duty. The training will include:

Training on operation and monitoring system of the wind farm

This is the type of training which are routinely carried out by the wind farm itself for new staffs.

Training on CDM basics with focus on monitoring

It will be carried out by the CDM consultancy company before the project is implemented. The CDM monitoring manual will be used as the primary training materials. The training includes the following contents:

- CDM project cycle and the significance of monitoring
- Management structure and work scope of each team member
- Components of the monitoring plan
- QA/QC procedure
- Monitoring report template
- Preparation for verification

5. Data Collection

The verification will use the main meter's data as long as the inaccuracy of the meter is within the permissible tolerance. The main procedures are as follows:

- I According to the requirements of power purchase/sales agreement, the project owner and the grid company should collect the two meters' data monthly, and check them at the same time.
- II The project owner supplies the electricity to the grid company, and provides an electricity sales receipts to the Grid Company. A copy of the sales receipts is stored by the project owner, together with a record of the payment by the grid company.
- III When the electricity generated by this project cannot meet the electricity requirement of the power plant, the grid company supplies the electricity to the project owner. The Grid Company provides



- an electricity sales receipts to the project owner and the sales receipts is stored by the project owner.
- IV The project owner records the power supplied to and purchased from the grid, and hence calculate the net electricity supplied to the grid;
 - V The project owner keeps and safe keeps the records of the main meter's data readings for verification by the DOE.

In case metering equipment is damaged and no reliable readings can be recorded the project entity will estimate net supply by the proposed project activity according to the following procedure:

In case metering equipment is damaged: The project entity and the grid company will jointly calculate a conservative estimate of power supplied to the grid. A statement will be prepared indicating

- I the background to the damage to metering equipment;
- II the assumptions used to estimate net supply to the grid for the days for which no record could be recorded;
- III the estimation of power supplied to the grid. The statement will be signed by both a representative of the project entity as well as a representative of the grid company.

The project entity will furthermore document all efforts taken to restore normal monitoring procedures.

6. Calibration of Meters & Metering

The metering equipment will be properly calibrated and checked according to the rules of Relative Technical Administrative Code of Electric Energy Metering (DL/T448-2000). The project owner will prepare backup procedures to deal with any errors occurred to the meters. The calibration records carried out by the grid company should be provided to the proposed project owner, and these records will be maintained by the proposed project owner and the third party designated.

Meters should be tested by a qualified metric organization co-authorized by the project owner and the grid company within 10 days after:

- I The detection of the reading difference between the main meter and the backup meter that exceeds the allowable tolerance.
- II The equipments malfunction caused by improper operation

All the calibration test records should be maintained safely for the verification.

7. Data Management System

To keep safely the record of the data collected during monitoring, this project will set up a complete data management system. The project will perfect the whole monitoring procedure by developing the CDM manual, tracking information from the primary source to the end-data calculations in paper document format. It is the responsibility of the proposed project owner to provide additional necessary data and information for validation and verification requirements of respective DOE. Physical documentation such as paper-based maps, diagrams and environmental assessment will be collated in a central place, together with this monitoring plan. All paper-based information will be stored by the proposed project owner and kept at least one copy.

At the end of each month, the monitoring data will be filed in a spreadsheet and stored on a hard disk, and the paper-based printout should be also archived. Furthermore, the project owner collects the



sales receipts for the electricity supplied to the grid as a cross-check, and compiled the monitoring report including the monitoring data and relevant evidence at the end of each crediting year.

All the data will be kept for two years following the end of the last crediting period.

8. Monitoring Report

After the CDM project manager collects and sorts the monitored data, the monitoring report is prepared by the project developer. The project developer has to make sure that the format and content of the monitoring report are consistent with the monitoring methodology in the registered PDD.

B.8. Date of completion of the application of the baseline study and monitoring methodology and the name of the responsible person(s)/entity(ies)
--

>>

Date of completion of baseline and monitoring methodology study 18/04/2012

Name of person/entity determining the baseline:

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Above persons and entity are not Project Participants.

**SECTION C. Duration of the project activity / crediting period****C.1 Duration of the project activity:****C.1.1. Starting date of the project activity:**

>>

30/03/2011(Turbines Purchasing contract signed date, which is considered as the earliest date of the proposed project)

C.1.2. Expected operational lifetime of the project activity:

>>

20 years 0 months

C.2 Choice of the crediting period and related information:**C.2.1. Renewable crediting period****C.2.1.1. Starting date of the first crediting period:**

>>

20/07/2012(or the date of registration whichever is later)

C.2.1.2. Length of the first crediting period:

>>

7 years 0 months

C.2.2. Fixed crediting period:**C.2.2.1. Starting date:**

>>

Not applicable

C.2.2.2. Length:

>>

Not applicable

**SECTION D. Environmental impacts****D.1. Documentation on the analysis of the environmental impacts, including transboundary impacts:**

>>

The environmental impact assessment for this project was completed in September 2010 and approved by Qingdao Environmental Protection Bureau on 12th October, 2010. The approval number is Qinghuanzi [2010] 209. A summary of the report is illustrated as below:

Ambient air

The impact on ambient air quality of the proposed project is mainly from dust during construction stage, by sprinkling water frequently and timely clearing can reduce the dust pollution. When the project is in operational period, there will be no air pollutions. In conclusion, the proposed project will not pose any threat on the quality of ambient air.

Impact from noise

There is some noise during the operation of wind turbines. The equipments and techniques with lower noise will be chosen to apply. Improvement on construction process and strengthening of equipment maintenance is emphasized. *The Standard of noise limits of construction site* (GB12523-90) should be fulfilled during the construction period and *Emission standard for industrial enterprise noise at boundary* (GB12348-2008) Level II noise standard would be fulfilled during operational period. Consequently, the noise of operation has little impact to the surrounding environment. Hence, the noise will not impact the work and daily life of local residents.

Electromagnetic impact

The operation of the wind farm will generate electromagnetic pollution, whereas the pollution is slight. In addition, the wind turbines are very far from local residents and village. Therefore, the electronic magnetic pollution to the surrounding environment is insignificant.

Impact from Solid waste

Solid wastes generated from the proposed project activity are excavated earth material and municipal solid waste. Part of the excavated earth material will be backfilled, and the rest will be used for land levelling and road construction near the project site. The municipal solid waste will be collected and treated together with the waste from local residents. As the report indicates, solid waste is handled properly.

Impact from Wastewater

Wastewater is mainly domestic wastewater. Wastewater quantity is fairly small and treatment methods will be applied for on-site primary treatment, and then the wastewater will be treated together with the local wastewater. Oxidation pond should be built on the site, through which the discharging wastewater would be fulfilled the *Reuse of recycling water for urban and Water quality standard for urban miscellaneous water consumption* (GB/T 18920-2002) will be used as virescence water. Therefore, the impact of wastewater is limited and mitigated.



The land where the proposed project located belongs to wasteland and no immigration is involved in the proposed project. Land compensation is carried out according the relative national regulations. No migrating birds have been found in the project field. Therefore, the project is not located on the passage of migrating birds, and the project construction will not influence the migration of birds.

D.2. If environmental impacts are considered significant by the project participants or the host Party, please provide conclusions and all references to support documentation of an environmental impact assessment undertaken in accordance with the procedures as required by the host Party:

>>

The environmental impacts of the Qingdao Longxin Wind Power Project Phase I are not considered as significant.

**SECTION E. Stakeholders' comments****E.1. Brief description how comments by local stakeholders have been invited and compiled:**

>>

In 20th Mar 2011, staff from the project owner carried out a survey of the local residents at Zhujiaying Village, Shuangmiao Village and Zhuangjiaying Village which belong to Liyuan Town and are around the project location. The staff introduced the background of the proposed project and then sent out 50 copies of questionnaire in a random way, 49 pieces of reply were received. Among the interviewees, 22 of them are farmers, 4 are workers, 13 are governmental officials, 7 are students and 3 others.

The questions regarding the proposed project were mainly as follows:

1. How is the current environment quality?

2. Are there any negative impacts of the proposed project on the everyday life of local residents?

3. Is the proposed project going to help improve the local economy?

4. Does the proposed project impact on local sound/noise environment?

5. Which is the environmental topic that concerns you the most during the construction and operation of the proposed project?

6. Do you support the proposed project?

E.2. Summary of the comments received:

>>

The summary of questionnaire survey is listed as the following:

- 37 (76%) of them think the current environment quality is good, 10 (20%) think it is normal and 2 (4%) of them are unsure;
- 38 (79%) of them think there will not be any negative impacts on their everyday life, 11 (27%) of them are not sure;
- 36 (73%) of them think the proposed project will help to improve the local economy, 13 (27%) of them are not sure;
- 20 (41%) of them think the proposed project doesn't impact on local sound/noise environment, 5 (10%) of them think it will, 27 (49%) of them are unsure;
- Regarding the construction and operation of the proposed project, 27 (55%) of them are most concerned with electromagnetic interference, 19 (40%) of them are most concerned with the noise level and 3 (6%) of them are most concerned with wastewater;
- 45 (92%) of them support the implementation of the proposed project and 4 (8%) of them don't care.

The summary of local stakeholders meeting:

The local community possesses basically positive comments on the effects of the proposed project. The interviewees considered that local social, economic and environmental development would be



beneficial from the proposed project. The response was overall supportive to the project implementation.

E.3. Report on how due account was taken of any comments received:

>>

The vast majority of public opinions received were supportive. A few of local residents were concerned about noise and electromagnetic interference during operation period. As discussed in the EIA report, these issues are well illuminated and specific instructions were given to mitigate the potential impacts of these issues. The project developer will control the noise strictly sourced from the project construction period and operation period according to the national regulations and standards. According to the EIA, the proposed project has little impact on the electromagnetic interference. In addition, the substation of the proposed project is designed to be about 500m enough far away from the residential area. Wastewater during the construction and operation phase will be removed periodically. The environmental impacts can be avoided, controlled or mitigated via thorough implementation of the mitigation measures. The local residents are satisfied with the treatment measures carried out by the project owner. Therefore, the proposed project can be carried out as planned.

**Annex 1****CONTACT INFORMATION ON PARTICIPANTS IN THE PROJECT ACTIVITY**

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Annex 2

INFORMATION REGARDING PUBLIC FUNDING

There is no public funding for the Qingdao Longxin Wind Power Project Phase I..

Annex 3

BASELINE INFORMATION

All the tables related to the calculation of baseline emission reduction are presented below:

Calculation of Operating Margin (OM)

Table A1. Simple OM Emission Factor of North China Power Grid in 2006

Fuel types	Unit	Beijing	Tianjin	Hebei	Shanxi	Inner Mongolia	Shandong	Subtotal	Carbon Content (t/TJ)	Oxidation rate (%)	Emission Coefficient of Fuel (kgCO ₂ /TJ)	Average low Caloric value (MJ/t, km ³)	CO ₂ emission (tCO ₂ e) L=G×J×K/10000 0 (unit of mass)
		A	B	C	D	E	F	G=A+B+C+D+E+F	H	I	J	K	L=G×J×K/10000 (unit of volume)
Raw coal	10000 ton	796.63	1639.2	6867.99	6968.88	8404.05	10930.66	35607.41	25.8	100	87,300	20,908	649,930,803
Cleaned coal	10000 ton						39.77	39.77	25.8	100	87,300	26,344	914,643
Other washed coal	10000 ton	6.36		214.13	371.14	61.77	544.6	1198	25.8	100	87,300	8,363	8,746,477
Briquette	10000 ton	7.97					27.77	35.74	26.6	100	87,300	20,908	652,351
Coke	10000 ton						3.23	3.23	29.2	100	95,700	28,435	87,896
Coke oven gas	10 ⁸ m ³	0.38	0.63	5.8	22.32	0.64	5.79	35.56	12.1	100	37,300	16,726	2,218,517
Other	10 ⁸ m ³	20.66	6.58	69.72	13.79	22.76	7.22	140.73	12.1	100	37,300	5,227	2,743,772



coal gas													
Crude oil	10000 ton					0.74		0.74	20	100	71,100	41,816	22,001
Gasoline	10000 ton			0.01				0.01	18.9	100	67,500	43,070	291
Diesel	10000 ton	0.21		3.01		0.07	6.32	9.61	20.2	100	72,600	42,652	297,577
Fuel oil	10000 ton	6.38		0.08			4.1	10.56	21.1	100	75,500	41,816	333,391
LPG	10000 ton						0.01	0.01	17.2	100	61,600	50,179	309
Refinery gas	10000 ton			2.43			2.32	4.75	15.7	100	48,200	46,055	105,443
Natural gas	10 ⁸ m ³	3.41	0.73		0.53			4.67	15.3	100	54,300	38,931	987,216
Other oil product	10000 ton						0.28	0.28	20	100	72,200	41,816	8,454
Other coking product	10000 ton							0	25.8	100	95,700	28,435	0
Other fuel	10000 tce	6.83		47.11	230.76	12.51	132.29	429.5	0	0	0	0	0
												Subtotal	667,049,139

Source: China Energy Statistical Yearbook 2007

TableA2. Thermal Power Generation of North China Power Grid in 2006

Province	Power Generation (MWh)	Ratio of Self Power Consumption of Plant (%)	Power Supply(MWh)
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Beijing	20,705,000	7.51	19,150,055
Tianjin	35,924,000	6.86	33,459,614
Hebei	143,888,000	6.63	134,348,226
Shanxi	150,250,000	7.45	139,056,375
Inner Mongolia	139,593,000	7.58	129,011,851
Shandong	230,922,000	7.12	214,480,354
Total	721,282,000		669,506,473

Source: China Electric Yearbook 2007

Table A3. Emission Factor of North China Power Grid in 2006

	Parameter	Unit	Value	Source
A	Net Import from Northeast China Power Grid to North China Power Grid (MWh)	MWh	2,618,060	China Power Industry Statistics 2007
B	Total Emissions of Northeast China Power Grid	tCO ₂ e	211,421,263	Calculation based on data from China Electric Power Yearbook and China Energy Statistical Yearbook
C	Total Power Supply of Northeast China Power Grid	MWh	183,890,005	China Electric Power Yearbook 2007
D	Average Emission Factor of Northeast China Power Grid	tCO ₂ e /MWh	1.14972	D=B/C
E	Net Import from Central China Power Grid to North China Power Grid (MWh)	MWh	497,060	China Power Industry Statistics 2007
F	Total Emissions of Central China Power Grid	tCO ₂ e	378,031,235	Calculation based on data from China Electric Power Yearbook and China Energy Statistical Yearbook



G	Total Power Supply of Central China Power Grid	MWh	337,056,176	China Electric Power Yearbook 2007
H	Average Emission Factor of Central China Power Grid	tCO ₂ e /MWh	1.12157	H=F/G
I	Total Power Supply of North China Power Grid	MWh	672,621,593	E=Total Power Generation of North China Power Grid + A+E
J	Total Emissions of North China Power Grid	tCO ₂ e	670,617,037	
K	Emission Factor of North China Power Grid	tCO ₂ e /MWh	0.99702	G=J/I

China Energy Statistical Yearbook 2007

Table A4. Simple OM Emission Factor of North China Power Grid in 2007

Fuel types	Unit	Beijing	Tianjin	Hebei	Shanxi	Inner Mongolia	Shandong	Subtotal	Carbon Content (t/TJ)	Oxidation rate (%)	Emission Coefficient of Fuel (kgCO ₂ /TJ)	Average low Caloric value (MJ/t, km ³)	CO ₂ emission (tCO ₂ e) L=G×J×K/100000 (unit of mass)
		A	B	C	D	E	F	G=A+B+C+D+E+F	H	I	J	K	L=G×J×K/10000 (unit of volume)



Raw coal	10000ton	816.17	1753.99	7716.13	7510.06	10434.25	11884.83	40115.43	25.8	100	87,300	20,908	732,214,267
Cleaned coal	10000ton						18.43	18.43	25.8	100	87,300	26,344	423,859
Other washed coal	10000ton	5.76		156.89	478.81	48.57	756.84	1446.87	25.8	100	87,300	8,363	10,563,452
Briquette	10000ton	7.93					42.86	50.79	26.6	100	87,300	20,908	927,054
Coke	10000ton			0.02			4.09	4.11	29.2	100	95,700	28,435	111,843
Coke oven gas	10 ⁸ m ³	0.07	0.72	3.13	25.46	2.58	13.61	45.57	12.1	100	37,300	16,726	2,843,020
Other coal gas	10 ⁸ m ³	11.8	7.6	88.38	72.8	28.17	29.64	238.39	12.1	100	37,300	5,227	4,647,821
Crude oil	10000ton							0	20	100	71,100	41,816	0
Gasoline	10000ton			0.01				0.01	18.9	100	67,500	43,070	291
Diesel	10000ton	0.33		2.35		0.62	5.08	8.38	20.2	100	72,600	42,652	259,490
Fuel oil	10000ton	4.74		0.18			2.35	7.27	21.1	100	75,500	41,816	229,522
LPG	10000ton							0	17.2	100	61,600	50,179	0
Refinery gas	10000ton	0.06		2.85			1.65	4.56	15.7	100	48,200	46,055	101,225
Natural gas	10 ⁸ m ³	5.03	0.73		0.54	4.22	0.01	10.53	15.3	100	54,300	38,931	2,225,993
Other oil product	10000ton	1.72						1.72	20	100	72,200	41,816	51,929
Other coking product	10000ton	4.74						4.74	25.8	100	95,700	28,435	128,986
Other fuel	10000 tce	11.94		77.25	360.26	30.75	163.48	643.68	0	0	0	0	0
												Subtotal	754,728,750



Source: China Energy Statistical Yearbook 2008

Table A5. Thermal Power Generation of North China Power Grid in 2007

Province	Power Generation (MWh)	Ratio of Self Power Consumption of Plant (%)	Power Supply(MWh)
Beijing	22,300,000	7.51	20,625,270
Tianjin	39,900,000	6.53	37,294,530
Hebei	163,300,000	6.67	152,407,890
Shanxi	173,400,000	7.99	159,545,340
Inner Mongolia	180,100,000	7.77	166,106,230
Shandong	259,100,000	7.23	240,367,070
Total	838,100,000		776,346,330

Source: China Electric Yearbook 2008

Table A6. Emission Factor of North China Power Grid in 2007

	Parameter	Unit	Value	Source
A	Net Import from Northeast China Power Grid to North China Power Grid	MWh	1,789,750	China Power Industry Statistics 2008
B	Total Emissions of Northeast China Power Grid	tCO ₂ e	219,122,791	Calculation based on data from China Electric Power Yearbook and China Energy Statistical Yearbook
C	Total Power Supply of Northeast China Power Grid	MWh	202,542,560	China Electric Power Yearbook 2008



D	Average Emission Factor of Northeast China Power Grid	tCO ₂ e /MWh	1.08186	D=B/C
E	Net Import from Central China Power Grid to North China Power Grid	MWh	803,000	China Power Industry Statistics 2008
F	Total Emissions of Central China Power Grid	tCO ₂ e	419,013,395	Calculation based on data from China Electric Power Yearbook and China Energy Statistical Yearbook
G	Total Power Supply of Central China Power Grid	MWh	380,239,080	China Electric Power Yearbook 2008
H	Average Emission Factor of Central China Power Grid	tCO ₂ e /MWh	1.10197	H=F/G
I	Total Power Supply of North China Power Grid	MWh	778,939,080	I=Total Power Generation of North China Power Grid + A + E
J	Total Emissions of North China Power Grid	tCO ₂ e	757,552,268	
K	Emission Factor of North China Power Grid	tCO ₂ e /MWh	0.97254	K=J/I

Table A7. Simple OM Emission Factor of North China Power Grid in 2008

Fuel types	Unit	Beijing	Tianjin	Hebei	Shanxi	Inner Mongolia	Shandong	Subtotal	Carbon Content	Oxidation rate	Emission Coefficient of Fuel	Average low Caloric value	CO ₂ emission (tCO ₂
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									(t/TJ)	(%)	(kgCO ₂ /TJ)	(MJ/t,k m3)	e) L=G×J×K/ 100000 (unit of mass)
		A	B	C	D	E	F	G=A+B+C+D+E+F	H	I	J	K	L=G×J×K/ 10000 (unit of volume)
Raw coal	10000ton	755.75	1,800.12	7,353.33	7,854.39	12,607.82	12,360.75	42,732.16	25.8	100	87300	20,908	779,976,613
Cleaned coal	10000 ton						23.88	23.88	25.8	100	87300	26,344	549,200
Other washed coal	10000 ton	5.05		134.52	582.39	66.2	691.21	1,479.37	25.8	100	87300	8,363	10,800,731
Briquette	10000 ton	5.66			32.49		45.38	83.53	26.6	100	87300	20,908	1,524,647
Coke	10000 ton			0.02			6.07	6.09	29.2	100	95700	28435	165,723
Coke oven gas	10 ⁸ m ³	0.11	0.86	8.37	24.55	3.55	16.2	53.64	12.1	100	37300	16,726	3,346,491
Other coal gas	10 ⁸ m ³	10.4	9.08	187.54	36	34.32	29.76	307.10	12.1	100	37300	5,227	5,987,440
Crude oil	10000 ton					0.02		0.02	20	100	71100	41,816	595
Gasoline	10000 ton							0.00	18.9	100	67500	43,070	0
Diesel	10000 ton	0.15		3.08		0.35		3.58	20.2	100	72600	42,652	110,856
Fuel oil	10000 ton	2.56		0.25				2.81	21.1	100	75500	41,816	88,715
LPG	10000 ton							0.00	17.2	100	61600	50,179	0
Refinery gas	10000 ton	0.44		2.93				3.37	15.7	100	48200	46,055	74,809



Natural gas	10 ⁸ m ³	11.09	0.7		0.97	2.12		14.88	15.3	100	54300	38,931	3,145,563
Other oil product	10000 ton	1.45						1.45	20	100	72200	41,816	43,777
Other coking product	10000 ton	7.97		7.61				15.58	25.8	100	95700	28,435	423,968
Other fuel	10000 tce	4.9	2.34	61.02	466	63.72	141.71	739.69	0	0	0	0	0
												Subtotal	806,239,126

Source: China Energy Statistical Yearbook 2009

Table A8. Thermal Power Generation of North China Power Grid in 2008

Province	Power Generation (MWh)	Ratio of Self Power Consumption of Plant (%)	Power Supply(MWh)
Beijing	24,300,000	7.14	22,564,980
Tianjin	39,700,000	7.05	36,901,150
Hebei	158,000,000	6.9	147,098,000
Shanxi	176,200,000	8.22	161,716,360
Inner Mongolia	200,800,000	7.96	184,816,320
Shandong	268,900,000	7.14	249,700,540
Total			802,797,350

Source: China Electric Yearbook 2009

Table A9. Emission Factor of North China Power Grid in 2008

	Parameter	Unit	Value	Source
A	Net Import from Northeast China Power Grid to North China	MWh	5,286,140	China Power Industry Statistics 2009



	Power Grid			
B	Total Emissions of Northeast China Power Grid	tCO ₂ e	231,611,818	Calculation based on data from China Electric Power Yearbook and China Energy Statistical Yearbook
C	Total Power Supply of Northeast China Power Grid	MWh	209,625,110	China Electric Power Yearbook 2009
D	Average Emission Factor of Northeast China Power Grid	tCO ₂ e /MWh	1.104886	D=B/C
E	Total Power Supply of North China Power Grid	MWh	808,083,490	E=Total Power Generation of North China Power Grid + A
F	Total Emissions of North China Power Grid	tCO ₂ e	812,079,707	
G	Emission Factor of North China Power Grid	MWh	1.00495	G=F/E

Table A10. Operating Margin Emission Factor of North China Power Grid

		Year 2006	Year 2007	Year 2008	Total
A	Emissions (tCO ₂ e /year)	670,616,651	757,549,895	812,079,707	2,240,246,253
B	Power Supply (MWh)	672,621,593	778,939,080	808,083,490	2,259,644,163
C	CO ₂ Emission Factor (tCO ₂ e/MWh)	C = A/B			0.99142



Calculation of Build Margin (BM):

Step 1. Calculation of weights of CO₂ emissions of solid, liquid and gas fuel in total emissions for power generation

$$\lambda_{Coal,y} = \frac{\sum_{i \in COAL,j} F_{i,j,y} \times NCV_{i,y} \times EF_{CO_2,i,y}}{\sum_{i,j} F_{i,j,y} \times NCV_{i,y} \times EF_{CO_2,i,y}}$$

$$\lambda_{Oil,y} = \frac{\sum_{i \in OIL,j} F_{i,j,y} \times NCV_{i,y} \times EF_{CO_2,i,y}}{\sum_{i,j} F_{i,j,y} \times NCV_{i,y} \times EF_{CO_2,i,y}}$$

$$\lambda_{Gas,y} = \frac{\sum_{i \in GAS,j} F_{i,j,y} \times NCV_{i,y} \times EF_{CO_2,i,y}}{\sum_{i,j} F_{i,j,y} \times NCV_{i,y} \times EF_{CO_2,i,y}}$$

Where:

$FC_{i,j,y}$: Amount of fossil fuel type i consumed in province j in year y (mass or volume unit, for solid and wet fuel, the unit is ton, for gas fuel, the unit is m³);

NCV_{ij} : Net thermal value of fuel type i in year y (for solid and wet fuel, the unit is GJ/t, for gas fuel, the unit is GJ/m³);

$EF_{CO_2,i,y}$: CO₂ emission factor of fossil fuel type i in year y (tCO₂ /GJ);

COAL, OIL and GAS respectively refers to the group of solid, liquid, and gas fuels.

Based on China Energy Statistical Yearbook 2009, the calculation of the weights of solid, liquid, and gas fuels in North China Power Grid are:

$$\lambda_{Coal} = 98.41\%, \quad \lambda_{Oil} = 0.03\%, \quad \lambda_{Gas} = 1.56\%$$

Step 2: Calculation of Emission Factor of Relevant Thermal Power



$$EF_{Thermal} = \lambda_{Coal} \times EF_{Coal, Adv} + \lambda_{Oil} \times EF_{Oil, Adv} + \lambda_{Gas} \times EF_{Gas, Adv}$$

Where: $EF_{Coal, Adv}$, $EF_{Oil, Adv}$ and $EF_{Gas, Adv}$ respectively refers to the emission factor representing best technology commercially available for fuel of coal, oil or gas fired power plants. For specific workings, see the following:

Table A11. Emission factor representing best technology commercially available for fuel of coal, oil or gas fired power plants

	Variable	Efficiency of Power Supply(%)	Emission Coefficient of Fuel (kgCO ₂ /TJ)	Oxidation Rate	Emission Factor (tCO ₂ e/MWh)
		A	B	C	D=3.6/A/1,000,000×B×C
Coal-fired Power Plant	$EF_{Coal, Adv}$	39.08	87,300	1	0.8042
Oil-fired Power Plant	$EF_{Oil, Adv}$	51.46	75,500	1	0.5282
Gas-fired Power Plant	$EF_{Gas, Adv}$	51.46	54,300	1	0.3799

$$EF_{Thermal} = \lambda_{Coal} \times EF_{Coal, Adv} + \lambda_{Oil} \times EF_{Oil, Adv} + \lambda_{Gas} \times EF_{Gas, Adv} = 0.7975 \text{ (tCO}_2\text{/MWh)}$$

Step 3: Calculation of BM of the Grid

$$EF_{grid, BM, y} = \frac{CAP_{Thermal, y}}{CAP_{Total, y}} \times EF_{Thermal, y}$$

Where: CAP_{Total} is the total of new capacity additions, and $CAP_{Thermal}$ is the new capacity addition of thermal power.

**Table A12. Installed Capacity of North China Power Grid in 2008**

Installed Capacity	Unit	Beijing	Tianjin	Hebei	Shanxi	Inner Mongolia	Shandong	Total
Thermal Power	MW	4,760.0	7,490.0	29,870.0	35,250.0	45,740.0	55,930.0	179,040.0
Hydropower	MW	1,050.0	0.0	1,540.0	790.0	830.0	1,050.0	5,260.0
Nuclear Power	MW	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Wind Power and Others	MW	0.0	0.0	700.0	0.0	2,300.0	370.0	3,370.0
Total	MW	5,810.0	7,490.0	32,110.0	36,040.0	48,860.0	57,350.0	187,660.0

Source: China Power Yearbook 2009

Table A13. Installed Capacity of North China Power Grid in 2007

Installed Capacity	Unit	Beijing	Tianjin	Hebei	Shanxi	Inner Mongolia	Shandong	Total
Thermal Power	MW	3,900	6,920	29,020	30,950	39,870	54,140	164,800
Hydropower	MW	1050	10	780	790	830	1,050	4,510
Nuclear Power	MW	0	0	0	0	0	0	0
Wind Power and Others	MW	2.7	0	410	0	1,096.5	210	1,719.2
Total		4,952.7	6,930	30,210	31,740	41,796.5	55,400	171,029.2

Source: China Power Yearbook 2008

**Table A14. Installed Capacity of North China Power Grid in 2006**

Installed Capacity	Unit	Beijing	Tianjin	Hebei	Shanxi	Inner Mongolia	Shandong	Total
Thermal Power	MW	3,984	6,512	26,087	26,661	28,899	49,395	141,538
Hydropower	MW	1,053	5	785	790	818	553	4,004
Nuclear Power	MW	0	0	0	0	0	0	0
Wind Power and Others	MW	24	24	218	0	565	106	937
Total	MW	5,061	6,541	27,090	27,451	30,282	50,054	146,479

Source: China Power Yearbook 2007

Table A15. Calculation of BM of North China Power Grid (MW)

	Installation in year 2006	Installation in year 2007	Installation in year 2008	New Additions from 2006 to 2008*	New Additions from 2007 to 2008*	Ratio in New Additions
	A	B	C	D	E	F
Thermal Power (MW)	141,538	164,800	179,040	46111	17,847	93.98%
Hydro Power (MW)	4,004	4,510	5,260	520	9	1.06%
Nuclear Power (MW)	0	0	0	0	0	0.00%
Wind Power and Others (MW)	937	1,719	3,370	2433	1,651	4.96%
Total (MW)	146,479	171,029	187,660	49,064	19,508	100.00%
Percentage compared with installation of 2008				26.15%	10.40%	

*Calculated with the consideration of installed, shut-down and pumped storage capacity

$$EF_{\text{grid,BM,y}} = 0.7975 \times 93.98\% = 0.7495 \text{ tCO}_2/\text{MWh}$$

**Table A16. Baseline Emission Factor of North China Power Grid**

	Parameter	Unit	Amount
A	Operating Margin Emission Factor	tCO ₂ /MWh	0.9914
B	Build Margin Emission Factor	tCO ₂ /MWh	0.7495
C	Combined Emission Factor ($C=0.75*A+0.25*B$)	tCO ₂ /MWh	0.9309

Table A17. Electricity Generation Baseline Emissions

	Parameter	Unit	Amount	Source or Equation
A	Project installed capacity	MW	49.8	Feasibility Study
B	Annual electricity supplied	MWh	99,026	Feasibility Study
C	Baseline Emissions Factor	tCO ₂ /MWh	0.9309	Table A16
D	Electricity generation baseline emissions	tCO ₂ /year	92,183	$D= B * C$



Annex 4

MONITORING INFORMATION

Please refer to B.7.2. in the PDD.